

Alpilla 600 MW Combined Cycle Power Plant

Tie-in and Terminal Point Register

62 points: 16 Owner/Contractor terminal points (TP-) under ALP-EPC-001 and 46 interfaces between the consortium members (IF-) under ALP-CA-001. 62 agreed or frozen, 0 draft. Checks: 0 warning(s), 21 note(s).

1. Conventions

- TP- points: Owner side / Contractor side under the EPC contract. IF- points: Istanbul EPC side / Imaginary Electric side under the consortium agreement. The scope break states who supplies the flange or weld end, counter-flange, gasket, bolts, the closing weld and its NDT/PWHT, supports and cable terminations; 'available by' is the date the providing side has its side ready, 'needed by' the date the receiving side needs it.
- Pressures in barg (vacuum as negative barg; HMB stream pressures in bar(a) are converted with 1.01325 bar), temperatures in degC. Design values are the mechanical design of the receiving connection.
- Operating envelope = the range the providing side guarantees at the point. It is checked against the HMB streams of all 9 HMB cases (MAX-NG-100, MIN-NG-100, SRC-LDO-100, SRC-NG-060, SRC-NG-100, SRC-NG-GT80, SRC-NG-MEL, SUM-NG-100, WIN-NG-100).
- Chain = point fed from an upstream point through passive equipment: the minimum upstream pressure less the agreed pressure loss (chain dp) must cover this point's minimum pressure; a higher upstream design pressure or temperature requires a declared protection (relief, slam-shut, safety valve set points).
- Status: draft (proposed), agreed (confirmed in writing by both sides, TQ reference), frozen (change only by TQ).

2. Overview

ID	Category	Service	Size / rating	Operating envelope	Available	Needed	Status
TP-A1	land access	Temporary construction area (5 ha)	-	-	2027-02-01	2027-03-01	agreed
TP-A2	land access	Site (plot) handover	-	-	2026-11-02	2026-11-16	agreed
TP-CW1	marine	Seawater intake head	about 2 x DN2200 (Contractor to size)	7-27 degC; max 40,000 m3/h	2027-04-30	2027-10-01	agreed
TP-CW2	marine	Outfall diffuser	-	12-34 degC; max 40,000 m3/h	2027-04-30	2027-10-01	agreed
TP-D1	civil	Storm water discharge	-	max 2.00 m3/s	2026-11-02	2027-01-15	agreed
TP-E1	electrical	Grid connection 380 kV	380 kV, 2 circuits	380.0 kV (342.0-420.0); max 640 MW	2028-07-15	2028-07-22	agreed
TP-E2	electrical	Construction power	34.5 kV / 5 MVA	34.5 kV (31.1-38.0); max 5.00 MVA	2027-02-28	2027-03-15	agreed
TP-G1	process	Natural gas supply	16 in (DN400), CL600	45.0-70.0 barg; 5-25 degC; max 115,000 Sm3/h	2028-06-30	2028-08-15	agreed
TP-L1	process	LDO supply	4 in tanker coupling (API RP 1004 bottom loading / dry-break)	0.0-3.0 barg; 0-40 degC; max 150 m3/h	2028-09-01	2028-11-01	agreed
TP-M1	marine	Temporary barge landing	-	-	2027-09-30	2027-11-01	agreed
TP-R1	land access	Road access	-	-	2026-11-02	2026-11-16	agreed
TP-S1	permit service	Sanitary wastewater	-	-	-	-	agreed
TP-T1	control	Telecommunication	-	-	2027-06-30	2028-05-01	agreed
TP-T2	control	Emission data link to the environmental authority	-	-	2028-08-01	2028-10-02	agreed
TP-T3	control	TSO SCADA and revenue metering link	-	-	2028-05-01	2028-06-01	agreed
TP-W1	utility	Potable water	DN150	2.5-6.0 barg; 5-25 degC; max 20.00 m3/h	2027-02-28	2027-03-15	agreed
IF-01	process	Natural gas to GT	12 in (DN300) CL600	36.5-41.0 barg; 5-40 degC; max 23.00 kg/s	2028-08-31	2028-09-18	agreed
IF-02	process	LDO to GT	6 in (DN150) CL150	3.0-6.0 barg; 10-45 degC; max 24.00 kg/s	2028-11-15	2028-12-01	agreed

ID	Category	Service	Size / rating	Operating envelope	Available	Needed	Status
IF-03	process	Demin water for water injection	4 in (DN100) CL150	3.0-6.0 barg; 10-40 degC; max 15.00 kg/s	2028-11-15	2028-12-01	agreed
IF-04	utility	Compressor wash water	2 in (DN50) CL150	3.0-6.0 barg; 10-40 degC; max 10.00 m3/h	2028-09-15	2028-12-15	agreed
IF-05	process	HP steam	12 in P91 (SA-335 P91)	59.0-192.0 barg; 560-600 degC; max 105 kg/s	2028-09-30	2028-10-16	agreed
IF-06	process	HP steam	12 in P91 (SA-335 P91)	52.0-186.0 barg; 560-600 degC; max 105 kg/s	2028-09-30	2028-10-16	agreed
IF-07	process	Cold reheat	24 in SA-335 P22	18.0-41.0 barg; 340-395 degC; max 105 kg/s	2028-09-30	2028-10-16	agreed
IF-08	process	Cold reheat	24 in SA-335 P22	17.0-41.0 barg; 340-395 degC; max 105 kg/s	2028-09-30	2028-10-16	agreed
IF-09	process	Hot reheat	28 in P91 (SA-335 P91)	17.0-38.5 barg; 550-600 degC; max 125 kg/s	2028-09-30	2028-10-16	agreed
IF-10	process	Hot reheat	28 in P91 (SA-335 P91)	15.5-38.5 barg; 550-600 degC; max 125 kg/s	2028-09-30	2028-10-16	agreed
IF-11	process	LP steam	20 in SA-106 C	1.5-5.0 barg; 210-260 degC; max 16.50 kg/s	2028-09-30	2028-10-16	agreed
IF-12	process	LP steam	20 in SA-106 C	1.1-5.0 barg; 210-260 degC; max 16.50 kg/s	2028-09-30	2028-10-16	agreed
IF-13	process	HP / IP / LP bypass valves	HP 12 in / IP 28 in / LP 20 in (per valve datasheets)	1.0-192.0 barg; 150-600 degC; max 125 kg/s	2028-02-01	2028-03-01	agreed
IF-14	process	Bypass steam to condenser	2 x 36 in + 1 x 28 in	-1.0-12.0 barg; 40-250 degC; max 142 kg/s	2028-05-01	2028-06-01	agreed
IF-15	process	Condensate from hotwell	20 in (DN500) CL150	-1.0-0.9 barg; 15-40 degC; max 140 kg/s	2028-09-01	2028-09-15	agreed
IF-16	process	Condensate to CPH	12 in (DN300) CL300	12.0-21.0 barg; 15-45 degC; max 150 kg/s	2028-09-01	2028-09-15	agreed
IF-17	process	BFP suction	16 in (DN400) CL150	1.5-6.0 barg; 130-165 degC; max 140 kg/s	2028-09-01	2028-09-15	agreed
IF-18	process	HP feedwater	10 in (DN250) CL2500	95.0-215.0 barg; 130-170 degC; max 110 kg/s	2028-09-01	2028-09-15	agreed
IF-19	process	IP feedwater	6 in (DN150) CL600	20.0-48.0 barg; 130-165 degC; max 33.00 kg/s	2028-09-01	2028-09-15	agreed
IF-20	process	FGH water return	4 in (DN100) CL600	18.0-46.0 barg; 60-90 degC; max 14.50 kg/s	2028-09-01	2028-09-15	agreed
IF-21	process	HRSG blowdown	2 in / 3 in	0.0-5.0 barg; 100-370 degC; max 5.00 kg/s	2028-09-01	2028-09-15	agreed
IF-22	process	HRSG drains and vents	various (1 in - 3 in)	0.0-1.0 barg; 40-610 degC; max 10.00 kg/s	2028-09-01	2028-09-15	agreed
IF-23	utility	Chemical dosing	1 in	3.0-215.0 barg; 5-40 degC; max 0.10 m3/h	2028-09-01	2028-09-15	agreed
IF-24	utility	Sampling	1/2 in	-1.0-215.0 barg; 20-600 degC; max 0.50 kg/s	2028-09-01	2028-09-15	agreed
IF-25	utility	Aqueous ammonia	1 in	3.0-7.0 barg; 5-40 degC; max 0.50 t/h	2028-11-01	2028-11-15	agreed
IF-26	process	Flue gas to stack	D 9.2 m	0.0-0.0 barg; 60-120 degC; max 850 kg/s	2028-06-30	2028-09-15	agreed
IF-27	process	Circulating water	4 x DN1800	0.5-3.0 barg; 5-35 degC; max 38,000 m3/h	2028-09-01	2028-09-15	agreed
IF-28	utility	Closed cooling water	various (DN50 - DN400)	3.0-6.0 barg; 20-36 degC; max 1,160 m3/h	2028-08-09	2028-08-16	agreed
IF-29	utility	Instrument and service air	1 in - 2 in	6.0-8.0 barg; 5-45 degC; max 500 Nm3/h	2028-08-09	2028-08-16	agreed
IF-30	utility	Nitrogen	1 in	5.0-8.0 barg; 5-40 degC; max 1,000 Nm3/h	2028-09-01	2028-09-15	agreed
IF-31	utility	Hydrogen / CO2	1/2 in	8.0-12.0 barg; 5-40 degC; max 60.00 Nm3/h	2028-08-25	2028-09-01	agreed
IF-32	utility	Auxiliary steam	4 in (DN100) CL300	8.0-9.5 barg; 250-300 degC; max 7.50 t/h	2028-09-30	2028-10-16	agreed
IF-33	control	Fire protection	-	-	2028-08-17	2028-08-24	agreed
IF-34	electrical	GT GSU HV	560 MVA	380.0 kV (342.0-420.0); max 560 MVA	2028-07-29	2028-08-02	agreed
IF-35	electrical	ST GSU HV	270 MVA	380.0 kV (342.0-420.0); max 270 MVA	2028-07-29	2028-08-02	agreed
IF-36	electrical	UAT supply	21 kV (GT) / 15.75 kV (ST)	21.0 kV (19.9-22.1); max 31.50 MVA	2028-08-09	2028-08-16	agreed
IF-37	electrical	MV supplies	10.5 kV feeders	10.5 kV (9.4-11.6); max 10.00 MVA	2028-08-09	2028-08-16	agreed
IF-38	electrical	LV supplies	0.4 kV feeders	0.4 kV (0.4-0.4)	2028-08-09	2028-08-16	agreed
IF-39	electrical	DC and UPS	220 V DC / 230 V AC UPS	0.2 kV (0.2-0.2)	2028-08-09	2028-08-16	agreed
IF-40	control	Control system interface	OPC UA + Modbus TCP, redundant; hardwired trips	-	2028-06-01	2028-06-15	agreed
IF-41	control	Protection interface	IEC 61850 + hardwired	-	2028-07-29	2028-08-02	agreed
IF-42	civil	Earthing	-	-	2027-10-15	2027-11-15	agreed

ID	Category	Service	Size / rating	Operating envelope	Available	Needed	Status
IF-43	civil	Foundations	-	-	2027-11-01	2027-11-15	agreed
IF-44	process	Condenser tube cleaning balls	2 x DN150 per inlet half	0.5-3.5 barg; 5-35 degC; max 200 m3/h	2028-09-01	2028-09-15	agreed
IF-45	process	HP attemperation spray water	3 in (DN80) CL2500	95.0-215.0 barg; 130-170 degC; max 5.00 kg/s	2028-09-01	2028-09-15	agreed
IF-46	process	Reheat attemperation spray water	2 in (DN50) CL600	20.0-48.0 barg; 130-170 degC; max 3.00 kg/s	2028-09-01	2028-09-15	agreed

3. Tie-in data cards

TP-A1 Temporary construction area (5 ha)

Contract / category	ALP-EPC-001 Engineering, Procurement and Construction Contract for the Alpilla 600 MW Combined Cycle Power Plant / land_access
Medium / location	land / E 0-180, N 320-600 (5.04 ha), north of the plot, west of the access road (E 90.0, N 460.0, EL +15.00 m)
Size / connection	- / Fenced 5.04 ha field E 0-180, N 320-600
Scope break	Owner: lease of the field until Taking-Over, free of charge to the Contractor. Contractor: fencing, surfacing, drainage, temporary facilities and restoration to the original condition.
Owner side	Lease of the 5 ha field until Taking-Over, free of charge to the Contractor
Contractor side	Laydown, pre-assembly, contractor facilities; restoration at the end
Isolation	-
Protection	-
Dates	available by 2027-02-01; needed by 2027-03-01
Agreement	agreed by Owner, Istanbul EPC in TQ-030 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-OWNR-0005, source:SRC-OWNR-0010
Remarks	Lease starts 2027-02-01 (TQ-030).
Checks (engine)	OK completeness: land_access: complete OK dates: available 2027-02-01, needed 2027-03-01: float 28 d

TP-A2 Site (plot) handover

Contract / category	ALP-EPC-001 Engineering, Procurement and Construction Contract for the Alpilla 600 MW Combined Cycle Power Plant / land_access
Medium / location	land / Plot E 0-400, N 0-300 (plant grid), levelled to +15.0 m (E 200.0, N 150.0, EL +15.00 m)
Size / connection	- / Plot boundary per land-use plan with boundary survey monuments
Scope break	Owner: plot per land-use plan, levelled to +15.0 m with compaction records, boundary monuments, free of encumbrances except the buried 11 kV utility cable along the east boundary (E 390, N 0-300), which the Owner relocates outside the plot by 2027-06-30; the east strip E 370-400 is not available to the Contractor before then. Contractor: everything within the plot boundary from the access date, incl. demolition of the abandoned foundations (E 300-360, N 20-60) and the 10 m channel buffer.
Owner side	Plot free of encumbrances, levelled to +15.0 m, boundary monuments; relocation of the buried 11 kV cable (E 390) by 2027-06-30
Contractor side	All works within the plot from the access date
Isolation	-
Protection	-
Conditions (text)	East strip E 370-400 available 2027-06-30 (needed by the Contractor 2027-09-01 for the gas station civil works).
Dates	available by 2026-11-02; needed by 2026-11-16
Scope items	SC-001
Agreement	agreed by Owner, Istanbul EPC in TQ-030 (2026-09-28)
Basis	source:SRC-OWNR-0005, source:SRC-OWNR-0010
Checks (engine)	OK completeness: land_access: complete OK dates: available 2026-11-02, needed 2026-11-16: float 14 d

TP-CW1 Seawater intake head

Contract / category	ALP-EPC-001 Engineering, Procurement and Construction Contract for the Alpilla 600 MW Combined Cycle Power Plant / marine
Medium / location	seawater / About 650 m offshore at -12 m (E 150.0, N -710.0, EL -12.00 m)
Size / connection	about 2 x DN2200 (Contractor to size) / Offshore intake head with velocity cap at -12 m
Scope break	Owner: seabed survey and marine permit for the intake location. Contractor: intake head (velocity cap <= 0.15 m/s), pipelines, screens and pump house (SC-016, SC-042).
Owner side	Seabed survey and marine permit
Contractor side	Intake head, pipelines, pump house
Isolation	-
Protection	-
Operating envelope	- to - barg; 7 to 27 degC
Flow	normal 36,050.00 / max 40,000.00 m3/h (density 1025.0 kg/m3)
HMB envelope (all cases)	stream 25 (flow temperature check): 10196.80-10196.80 kg/s, 1.487-1.487 barg, 6.9-27.0 degC
Quality / specification	source:SRC-OWNR-0002, requirement:ER-02.04

Conditions (text)	6.9-27.0 degC
Dates	available by 2027-04-30; needed by 2027-10-01
Scope items	SC-016
Agreement	agreed by Owner, Istanbul EPC in TQ-030 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-OWNR-0005, source:SRC-OWNR-0010
Remarks	Capacity includes about 2,000 m3/h for SWRO and electro-chlorination in addition to the 38,000 m3/h CW design flow; pressure at the intake is hydrostatic (HMB pressure check not applicable).
Checks (engine)	OK completeness: marine: complete OK HMB flow: stream 25 max 10196.80 kg/s (WIN-NG-100) <= capacity 11388.89 kg/s (margin +10.5 %), 9 cases OK HMB temperature: stream 25: 6.9 - 27.0 degC within 6.9 - 27.0 degC (design None) INFO HMB scope: stream 25: flow + temperature checked only - see remarks OK dates: available 2027-04-30, needed 2027-10-01: float 154 d

TP-CW2 Outfall diffuser

Contract / category	ALP-EPC-001 Engineering, Procurement and Construction Contract for the Alpilla 600 MW Combined Cycle Power Plant / marine
Medium / location	seawater / About 350 m offshore, at least 250 m from the intake (E 260.0, N -410.0, EL -8.00 m)
Size / connection	- / Outfall diffuser about 350 m offshore at -8 m
Scope break	Owner: marine permit and discharge permit (EIA conditions). Contractor: outfall pipeline and diffuser; treated effluents (SC-020, SC-057) join the outfall upstream of the discharge monitoring.
Owner side	Marine permit
Contractor side	Outfall pipeline and diffuser
Isolation	-
Protection	-
Operating envelope	- to - barg; 12 to 34 degC
Flow	normal 36,050.00 / max 40,000.00 m3/h (density 1025.0 kg/m3)
HMB envelope (all cases)	stream 26 (flow temperature check): 10196.80-10196.80 kg/s, 0.687-0.687 barg, 14.1-33.4 degC
Quality / specification	requirement:ER-08.04, requirement:ER-08.05, source:SRC-OWNR-0007
Dates	available by 2027-04-30; needed by 2027-10-01
Scope items	SC-016, SC-020, SC-057
Agreement	agreed by Owner, Istanbul EPC in TQ-030 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-OWNR-0005, source:SRC-OWNR-0010
Remarks	Discharge limits: <= 3 K excess at the edge of the 100 m mixing zone, residual chlorine <= 0.1 mg/l; pressure at the diffuser is hydrostatic (HMB pressure check not applicable).
Checks (engine)	OK completeness: marine: complete OK HMB flow: stream 26 max 10196.80 kg/s (WIN-NG-100) <= capacity 11388.89 kg/s (margin +10.5 %), 9 cases OK HMB temperature: stream 26: 14.1 - 33.4 degC within 12.0 - 34.5 degC (design None) INFO HMB scope: stream 26: flow + temperature checked only - see remarks OK dates: available 2027-04-30, needed 2027-10-01: float 154 d

TP-D1 Storm water discharge

Contract / category	ALP-EPC-001 Engineering, Procurement and Construction Contract for the Alpilla 600 MW Combined Cycle Power Plant / civil
Medium / location	storm water / Western stream channel (max 2 m3/s) or to sea (E 0.0, N 150.0, EL +13.00 m)
Size / connection	- / Contractor outlet structure with energy dissipation at the western stream channel
Scope break	Owner: water authority consent for the discharge. Contractor: separate clean/oily drainage, oil interceptors (ER-09.06), outlet structure on the plot boundary and the channel protection within 5 m of the outlet.
Owner side	Water authority consent
Contractor side	Site drainage, oil interceptors, outlet structure
Isolation	-
Protection	-
Flow	normal - / max 2.00 m3/s (density 1000.0 kg/m3)
Quality / specification	requirement:ER-09.06, source:SRC-OWNR-0007
Dates	available by 2026-11-02; needed by 2027-01-15
Agreement	agreed by Owner, Istanbul EPC in TQ-030 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-OWNR-0005, source:SRC-OWNR-0010
Checks (engine)	OK completeness: civil: complete OK dates: available 2026-11-02, needed 2027-01-15: float 74 d

TP-E1 Grid connection 380 kV

Contract / category	ALP-EPC-001 Engineering, Procurement and Construction Contract for the Alpilla 600 MW Combined Cycle Power Plant / electrical
Medium / location	electricity / Line gantry at the north-west corner (E 30.0, N 290.0, EL +15.00 m)
Size / connection	380 kV, 2 circuits / Line-side terminals of the Contractor's tension insulator strings on the gantry
Scope break	Owner (line contractor): OHL, conductors and OPGW strung to the tension insulator strings on the Contractor's line gantry, incl. the line clamps. Contractor: gantry, insulator strings and hardware, jumpers and downleads to the GIS outdoor bushings, surge arresters, OPGW joint box on the gantry and fibre to the telecom room.
Owner side	Double-circuit 380 kV OHL up to the dead-end tower, conductors to the gantry

Contractor side	Gantry, GIS switchyard, protection and metering
Isolation	Contractor GIS line bay circuit breaker, line disconnecter and line earthing switch; Owner/TSO breaker at the remote end (Kiyi substation).
Protection	Export limiter in the plant control system: 640 MW net at the 380 kV revenue meters (TQ-027); line differential 87L main 1 / main 2 with the remote end (plant end by the Contractor, remote end by the Owner), settings coordinated with the TSO (ER-18.02).
Metering	Revenue metering class 0.2S on both line bays (SC-035), TSO approved.
Flow	normal - / max 640.00 MW - actively limited
Electrical	380.00 kV (342.0-420.0 kV); Isc 50.0 kA, rating 50.0 kA
HMB power	net
Quality / specification	source:SRC-OWNR-0006, requirement:ER-11.01
Conditions (text)	380 kV nominal, 420 kV highest; 360-410 kV normal, 342-420 kV up to 30 min; 49.0-51.0 Hz; short-circuit 12-40 kA (2029), 50 kA (2035 planning); connection capacity 640 MW export / 40 MW import (ALP-OWN-ELE-001).
Dates	available by 2028-07-15; needed by 2028-07-22
Scope items	SC-030, SC-031, SC-048
Agreement	agreed by Owner, Istanbul EPC in TQ-030 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-OWNR-0005, source:SRC-OWNR-0009, source:SRC-OWNR-0010
Remarks	Short-circuit values include the Alpilla generators' contribution (TSO, TQ-031). Reactive capability per ER-04.08 at the generator terminals (TQ-032).
Checks (engine)	OK completeness: electrical: complete OK short circuit: prospective 50.0 kA <= rating 50.0 kA INFO HMB power: max net 665.7 MW (WIN-NG-100) > 640 MW (margin -4.0 %); limited to 640 MW at this point: Export limiter in the plant control system: 640 MW net at the 380 kV revenue meters (TQ-027); line differential 87L main 1 / main 2 with the remote end (plant end by the Contractor, remote end by the Owner), settings coordinated with the TSO (ER-18.02). INFO dates: available 2028-07-15, needed 2028-07-22: float 7 d (tight)

TP-E2 Construction power

Contract / category	ALP-EPC-001 Engineering, Procurement and Construction Contract for the Alpilla 600 MW Combined Cycle Power Plant / electrical
Medium / location	electricity / North gate (E 200.0, N 300.0, EL +15.00 m)
Size / connection	34.5 kV / 5 MVA / 34.5 kV cable terminations at the outgoing side of the utility metering kiosk
Scope break	Owner (distribution company): 34.5 kV line/cable to the metering kiosk at the north gate, incl. the utility meter and load-break switch. Contractor: cable from the kiosk outgoing terminals, 34.5 kV ring main unit and construction substations.
Owner side	34.5 kV connection, 5 MVA, metering
Contractor side	Construction substations and distribution
Isolation	Utility load-break switch in the kiosk; Contractor RMU incoming switch with earthing switch.
Protection	Contractor RMU incoming circuit breaker (overcurrent / earth fault) graded with the utility.
Metering	Utility meter in the kiosk (construction energy paid by the Contractor, SC-045).
Flow	normal - / max 5.00 MVA
Electrical	34.50 kV (31.1-38.0 kV); Isc 12.5 kA, rating 16.0 kA
Conditions (text)	34.5 kV +/- 10 %, 5 MVA, prospective short-circuit 12.5 kA (distribution company).
Dates	available by 2027-02-28; needed by 2027-03-15
Scope items	SC-045, SC-045
Agreement	agreed by Owner, Istanbul EPC in TQ-030 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-OWNR-0005, source:SRC-OWNR-0009, source:SRC-OWNR-0010
Checks (engine)	OK completeness: electrical: complete OK short circuit: prospective 12.5 kA <= rating 16.0 kA OK dates: available 2027-02-28, needed 2027-03-15: float 15 d

TP-G1 Natural gas supply

Contract / category	ALP-EPC-001 Engineering, Procurement and Construction Contract for the Alpilla 600 MW Combined Cycle Power Plant / process
Medium / location	natural gas / North-east plot corner, flange at EL +14.0 (E 380.0, N 300.0, EL +14.00 m)
Size / connection	16 in (DN400), CL600 / ASME B16.5 CL600 RF weld-neck flange, 16 in
Scope break	Owner: spur line, tap valve station, remotely operable isolation valve and monolithic insulating joint; scope break at the downstream flange of the insulating joint. Contractor: counter-flange, gasket, stud bolts and everything downstream (gas receiving and metering station, SC-028). The Owner's cathodic protection ends at the insulating joint; plant earthing starts downstream.
Owner side	16-inch spur line (4.5 km), tap valve station, isolation valve and insulating joint
Contractor side	Gas receiving and metering station from the inlet flange
Isolation	Owner boundary isolation valve (operated by the network operator); Contractor station inlet ESD valve.
Protection	Owner pipeline MAOP 75 barg; the Contractor's station is designed for 75 barg up to the regulators.
Metering	Fiscal metering by the Contractor in the station, approved by the network operator (ER-18.03).
Design	75.0 barg / 50 degC
Operating envelope	45.0 to 70.0 barg; 5 to 25 degC
Flow	normal 101,600.00 / max 115,000.00 Sm ³ /h (density 0.7325 kg/m ³)
Quality / specification	source:SRC-OWNR-0003, requirement:ER-03.06
Conditions (text)	Normal 55-65 barg; minimum 45 barg for full load; exceptional 40 barg up to 72 h/year (GT load limited, ER-03.03); MOP 70 / MAOP 75 barg; 5-25 degC; 115,000 Sm ³ /h firm (density 0.7325 kg/Sm ³ at 15 degC, reference gas).

Dates	available by 2028-06-30; needed by 2028-08-15
Scope items	SC-028, SC-027, SC-047
Agreement	agreed by Owner, Istanbul EPC in TQ-030 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-OWNR-0005, source:SRC-OWNR-0009, source:SRC-OWNR-0010
Checks (engine)	OK completeness: process: complete OK dates: available 2028-06-30, needed 2028-08-15: float 46 d

TP-L1 LDO supply

Contract / category	ALP-EPC-001 Engineering, Procurement and Construction Contract for the Alpilla 600 MW Combined Cycle Power Plant / process
Medium / location	light diesel oil / North gate (road tankers) (E 200.0, N 300.0, EL +15.00 m)
Size / connection	4 in tanker coupling (API RP 1004 bottom loading / dry-break) / Dry-break tanker coupling at the Contractor's unloading bays
Scope break	Owner (LDO supplier): road tankers delivered to the unloading bays, tanker hoses and coupling halves on the tanker, first fill of the storage tanks (SC-055). Contractor: unloading bays, coupling halves on the plant side, pumps, metering, storage and treatment.
Owner side	LDO supply contract (up to 25 trucks/day), first fill
Contractor side	Unloading station and storage
Isolation	Tanker bottom valve (supplier); Contractor unloading valve and ESD at each bay.
Protection	Contractor unloading pumps relief; bay earthing and overfill interlock (tanker earthing permissive).
Metering	Contractor unloading flow meter per bay; tanker delivery notes (Owner supplier).
Design	10.0 barg / 50 degC
Operating envelope	0.000 to 3.0 barg; 0 to 40 degC
Flow	normal 100.00 / max 150.00 m3/h (density 835.0 kg/m3)
Quality / specification	source:SRC-OWNR-0004, requirement:ER-03.07
Conditions (text)	Up to 25 trucks/day (about 30 m3 each); 4 bays at about 100 m3/h each unloading pump set; LDO per TS EN 590 based specification.
Dates	available by 2028-09-01; needed by 2028-11-01
Scope items	SC-052, SC-055, SC-052
Agreement	agreed by Owner, Istanbul EPC in TQ-030 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-OWNR-0005, source:SRC-OWNR-0010
Checks (engine)	OK completeness: process: complete OK dates: available 2028-09-01, needed 2028-11-01: float 61 d

TP-M1 Temporary barge landing

Contract / category	ALP-EPC-001 Engineering, Procurement and Construction Contract for the Alpilla 600 MW Combined Cycle Power Plant / marine
Medium / location	heavy transport / Public coastal strip south of the plot (E 120.0, N -40.0, EL +1.50 m)
Size / connection	- / Temporary ro-ro barge landing on the public coastal strip
Scope break	Owner: coastal permit (application by 2027-04-30 based on the Contractor's design). Contractor: design, construction, operation, removal and reinstatement of the barge landing; heavy haul road to the site.
Owner side	Coastal permit: application by 2027-04-30 if the Contractor's permit design is received by 2027-03-31; permit targeted by 2027-09-30
Contractor side	Design, construction, operation and removal of the landing; marine works only 1 October - 30 April
Isolation	-
Protection	-
Conditions (text)	Heaviest pieces (IEC-ALP-REQ-001 section 10): GT core engine 405 t (13.2 x 5.9 x 5.6 m, about 480 t gross on SPMT), GT generator stator 340 t, GT GSU 268 t; landing and haul road designed for 480 t gross SPMT loads.
Dates	available by 2027-09-30; needed by 2027-11-01
Agreement	agreed by Owner, Istanbul EPC in TQ-030 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-OWNR-0005, source:SRC-OWNR-0007, source:SRC-OWNR-0009, source:SRC-OWNR-0010
Checks (engine)	OK completeness: marine: complete OK dates: available 2027-09-30, needed 2027-11-01: float 32 d

TP-R1 Road access

Contract / category	ALP-EPC-001 Engineering, Procurement and Construction Contract for the Alpilla 600 MW Combined Cycle Power Plant / land access
Medium / location	road / North gate (E 200.0, N 300.0, EL +15.00 m)
Size / connection	- / Existing 6 m asphalt road at the north gate (E 200, N 300)
Scope break	Owner: existing road to the state road up to the north gate. Contractor: site roads from the gate, road condition survey before and after construction and repair of construction damage; abnormal loads via TP-M1.
Owner side	Existing 6 m asphalt road to the state road
Contractor side	Site roads and gate
Isolation	-
Protection	-
Conditions (text)	6 m asphalt, state road bridges limited to 120 t gross vehicle weight; components above about 90 t via the barge landing TP-M1 (ALP-OWN-GEN-001 section 3).

Dates	available by 2026-11-02; needed by 2026-11-16
Agreement	agreed by Owner, Istanbul EPC in TQ-030 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-OWNR-0005, source:SRC-OWNR-0009, source:SRC-OWNR-0010
Checks (engine)	OK completeness: land_access: complete OK dates: available 2026-11-02, needed 2026-11-16: float 14 d

TP-S1 Sanitary wastewater

Contract / category	ALP-EPC-001 Engineering, Procurement and Construction Contract for the Alpilla 600 MW Combined Cycle Power Plant / permit_service
Medium / location	sewage / None - no municipal sewer available
Size / connection	- / -
Scope break	No physical connection (no municipal sewer). The Contractor's packaged sanitary treatment plant (SC-057) discharges treated effluent into the outfall (TP-CW2) within the Owner's discharge permit.
Owner side	Nothing (no sewer)
Contractor side	Packaged sanitary treatment plant (SC-057); treated effluent to the outfall
Isolation	-
Protection	-
Quality / specification	source:SRC-OWNR-0007
Dates	available by -; needed by -
Scope items	SC-057, SC-020
Agreement	agreed by Owner, Istanbul EPC in TQ-030 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-OWNR-0005, source:SRC-OWNR-0010
Checks (engine)	OK completeness: permit_service: complete

TP-T1 Telecommunication

Contract / category	ALP-EPC-001 Engineering, Procurement and Construction Contract for the Alpilla 600 MW Combined Cycle Power Plant / control
Medium / location	fibre / North gate (E 200.0, N 300.0, EL +14.00 m)
Size / connection	- / 2 x HDPE ducts with draw pits at the plot boundary (two operators)
Scope break	Owner: operators' fibre in ducts up to the boundary draw pits. Contractor: ducts and fibre from the draw pits to the telecom room, telecom equipment.
Owner side	Fibre ducts of two operators to the plot boundary
Contractor side	Site telecom network and SCADA links
Isolation	-
Protection	-
Dates	available by 2027-06-30; needed by 2028-05-01
Scope items	SC-035, SC-044
Agreement	agreed by Owner, Istanbul EPC in TQ-030 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-OWNR-0005, source:SRC-OWNR-0009, source:SRC-OWNR-0010
Checks (engine)	OK completeness: control: complete OK dates: available 2027-06-30, needed 2028-05-01: float 306 d

TP-T2 Emission data link to the environmental authority

Contract / category	ALP-EPC-001 Engineering, Procurement and Construction Contract for the Alpilla 600 MW Combined Cycle Power Plant / control
Medium / location	data / CEMS data logger in the control building
Size / connection	- / Authority continuous emission monitoring protocol over the public network (VPN), per the Ministry's data-link rules
Scope break	Owner: registration with the Ministry of Environment continuous emission monitoring centre, the operator credentials and the data-link agreement. Contractor: CEMS (SC-038), data acquisition system, firewall and VPN router in the control building; scope break at the router's public network port.
Owner side	Registration of the plant with the Ministry's continuous emission monitoring centre, credentials, connection agreement
Contractor side	CEMS, data acquisition and the data link (SC-038) up to the Ministry server interface
Isolation	-
Protection	-
Dates	available by 2028-08-01; needed by 2028-10-02
Scope items	SC-038
Agreement	agreed by Owner, Istanbul EPC in TQ-030 (2026-09-28)
Basis	source:SRC-OWNR-0005, source:SRC-OWNR-0010
Checks (engine)	OK completeness: control: complete OK dates: available 2028-08-01, needed 2028-10-02: float 62 d

TP-T3 TSO SCADA and revenue metering link

Contract / category	ALP-EPC-001 Engineering, Procurement and Construction Contract for the Alpilla 600 MW Combined Cycle Power Plant / control
Medium / location	data / TSO RTU and metering cubicle in the control building
Size / connection	- / IEC 60870-5-104 over OPGW fibre (TP-E1) and a second path via the operator fibre (TP-T1)

Scope break	Owner/TSO: signal list, IEC 60870-5-104 profile, TSO end equipment and the OPGW up to the joint box on the Contractor's gantry. Contractor: joint box, fibre to the telecom room, RTU, revenue meters and SCADA tests with the TSO.
Owner side	TSO signal list and protocol, TSO end of the link, OPGW on the Owner's line
Contractor side	RTU, revenue meters, fibre from the gantry joint box, telecom equipment (SC-035)
Isolation	-
Protection	-
Dates	available by 2028-05-01; needed by 2028-06-01
Scope items	SC-035
Agreement	agreed by Owner, Istanbul EPC in TQ-030 (2026-09-28)
Basis	source:SRC-OWNR-0005, source:SRC-OWNR-0010
Checks (engine)	OK completeness: control: complete OK dates: available 2028-05-01, needed 2028-06-01: float 31 d

TP-W1 Potable water

Contract / category	ALP-EPC-001 Engineering, Procurement and Construction Contract for the Alpilla 600 MW Combined Cycle Power Plant / utility
Medium / location	water / North gate (E 200.0, N 300.0, EL +14.00 m)
Size / connection	DN150 / DN150 PN16 EN 1092-1 flange at the outlet of the Owner's meter pit valve
Scope break	Owner: municipal main, connection, meter, backflow preventer and outlet valve in the meter pit at the north gate. Contractor: from the outlet flange (counter-flange, gasket, bolts) to storage and distribution.
Owner side	DN150 municipal main connection and meter, 3 barg, 20 m3/h
Contractor side	Potable water storage and distribution
Isolation	Owner valves before and after the meter; Contractor isolation valve downstream.
Protection	Owner backflow preventer (TS EN 1717).
Metering	Municipal meter in the pit.
Design	10.0 barg / 30 degC
Operating envelope	2.5 to 6.0 barg; 5 to 25 degC
Flow	normal 10.00 / max 20.00 m3/h (density 1000.0 kg/m3)
Quality / specification	source:SRC-OWNR-0005
Conditions (text)	2.5-6.0 barg at the meter outlet, 20 m3/h (municipal supply), potable quality per the Turkish drinking water regulation.
Dates	available by 2027-02-28; needed by 2027-03-15
Scope items	SC-043, SC-045
Agreement	agreed by Owner, Istanbul EPC in TQ-030 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-OWNR-0005, source:SRC-OWNR-0009, source:SRC-OWNR-0010
Checks (engine)	OK completeness: utility: complete OK dates: available 2027-02-28, needed 2027-03-15: float 15 d

IF-01 Natural gas to GT

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / process
Medium / location	natural gas / performance gas heater inlet flange
Size / connection	12 in (DN300) CL600 / ASME B16.5 CL600 RF weld-neck flange
Scope break	IEC supplies the performance gas heater inlet flange; EPC supplies the counter-flange, gasket (spiral wound, 316L/graphite), stud bolts and the pipe support nearest the flange, and makes the joint. Hydrotest up to the flange by each party; joint leak test with gas-in by EPC.
Istanbul EPC side	EPC: gas from EPC gas receiving/metering station
Imaginary Electric side	IEC: performance gas heater inlet flange
Isolation	EPC: gas station outlet ESD valve (fail-closed, SIL 2, closed by the TCS gas trip) and manual valve with spectacle blind upstream of IF-01. IEC: gas skid stop and control valves.
Protection	EPC gas station pressure control: monitor + active regulator, slam-shut set 46 barg, token relief 44 barg; downstream design 63 barg (EN 12186 / ASME B31.8).
Metering	Fiscal metering (ultrasonic 2 x 100 % + gas chromatograph) in the EPC gas station; IEC skid flow meter for control only.
Design	63.0 barg / 80 degC
Operating envelope	36.5 to 41.0 barg; 5 to 40 degC
Flow	normal 20.67 / max 23.00 kg/s
HMB envelope (all cases)	stream 2 (full check): 8.68-22.36 kg/s, 36.987-36.987 barg, 25.0-25.0 degC
Fed from	TP-G1 (chain dp 4.7 bar)
Quality / specification	requirement:ER-03.06, source:SRC-OWNR-0003, source:SRC-EPCE-0010
Conditions (text)	Pressure rate of change <= 0.7 bar/s; >= 28 K superheat above hydrocarbon and water dew points; filtration 3 um absolute; below 36.5 barg the TCS load limiter applies (DEC-EPCE-0003).
Dates	available by 2028-08-31; needed by 2028-09-18
Scope items	SC-029
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0005, source:SRC-IECE-0008

Checks (engine)	OK completeness: process: complete OK HMB flow: stream 2 max 22.36 kg/s (WIN-NG-100) <= capacity 23.00 kg/s (margin +2.8 %), 8 cases OK HMB pressure: stream 2: 36.987 - 36.987 barg within 36.5 - 41.0 barg (design 63.0) OK HMB temperature: stream 2: 25.0 - 25.0 degC within 5.0 - 40.0 degC (design 80.0) OK chain pressure: TP-G1 min 45.0 barg - 4.7 bar = 40.30 barg >= required 36.5 barg OK chain design: TP-G1 design 75 barg > 63.0 barg here: protected - EPC gas station pressure control: monitor + active regulator, slam-shut set 46 barg, token relief 44 barg; downstream de OK chain flow: capacity here 23.00 kg/s <= TP-G1 23.40 kg/s OK dates: available 2028-08-31, needed 2028-09-18: float 18 d
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IF-02 LDO to GT

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / process
Medium / location	LDO / liquid fuel skid inlet flange
Size / connection	6 in (DN150) CL150 / ASME B16.5 CL150 RF flange
Scope break	IEC supplies the liquid fuel skid inlet flange; EPC supplies counter-flange, gasket, bolts, the last pipe support and the joint.
Istanbul EPC side	EPC: LDO forwarding/treatment system
Imaginary Electric side	IEC: liquid fuel skid inlet flange
Isolation	EPC: LDO ESD valve (fail-closed) and manual valve with spectacle blind at the skid boundary. IEC: skid stop valve.
Protection	EPC forwarding pumps: discharge relief to tank set 12 barg (< 16 barg design).
Metering	EPC Coriolis meter on the forwarding skid (consumption and performance tests).
Design	16.0 barg / 60 degC
Operating envelope	3.0 to 6.0 barg; 10 to 45 degC
Flow	normal 22.41 / max 24.00 kg/s
HMB envelope (all cases)	stream 27 (full check): 22.41-22.41 kg/s, 3.987-3.987 barg, 30.0-30.0 degC
Quality / specification	requirement:ER-03.07, source:SRC-OWNR-0004, source:SRC-EPCE-0010
Conditions (text)	LDO treated by the EPC to the GT inlet limits (ER-03.07); filtration 5 um.
Dates	available by 2028-11-15; needed by 2028-12-01
Scope items	SC-053, SC-054
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0005, source:SRC-IECE-0008
Checks (engine)	OK completeness: process: complete OK HMB flow: stream 27 max 22.41 kg/s (SRC-LDO-100) <= capacity 24.00 kg/s (margin +6.6 %), 1 cases OK HMB pressure: stream 27: 3.987 - 3.987 barg within 3.0 - 6.0 barg (design 16.0) OK HMB temperature: stream 27: 30.0 - 30.0 degC within 10.0 - 45.0 degC (design 60.0) OK dates: available 2028-11-15, needed 2028-12-01: float 16 d

IF-03 Demin water for water injection

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / process
Medium / location	demin water / water injection skid inlet
Size / connection	4 in (DN100) CL150 / ASME B16.5 CL150 RF flange
Scope break	IEC supplies the water injection skid inlet flange (skid with high-pressure pumps by IEC); EPC supplies counter-flange, gasket, bolts and the demin water forwarding pumps.
Istanbul EPC side	EPC: demin water storage/pumps
Imaginary Electric side	IEC: water injection skid inlet
Isolation	EPC manual isolation valve and spectacle blind; IEC skid inlet valve.
Protection	EPC demin forwarding pumps shut-off head <= 12 barg (< 16 barg design).
Design	16.0 barg / 50 degC
Operating envelope	3.0 to 6.0 barg; 10 to 40 degC
Flow	normal 13.44 / max 15.00 kg/s
HMB envelope (all cases)	stream 28 (flow temperature check): 13.44-13.44 kg/s, 58.987-58.987 barg, 30.0-30.0 degC
Quality / specification	requirement:ER-09.03, source:SRC-EPCE-0010
Conditions (text)	Water/fuel ratio 0.60 on LDO (TQ-IEC-018). HMB stream 28 is at the IEC skid discharge (60 bar(a)); pressure at IF-03 is 3.0-6.0 barg.
Dates	available by 2028-11-15; needed by 2028-12-01
Scope items	SC-054
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0005, source:SRC-IECE-0008
Remarks	HMB pressure check not applicable: PER-003 stream 28 is downstream of the IEC injection pumps (TQ-IEC-021 item 3); the injection pump power is part of the IEC GT auxiliaries.
Checks (engine)	OK completeness: process: complete OK HMB flow: stream 28 max 13.44 kg/s (SRC-LDO-100) <= capacity 15.00 kg/s (margin +10.4 %), 1 cases OK HMB temperature: stream 28: 30.0 - 30.0 degC within 10.0 - 40.0 degC (design 50.0) INFO HMB scope: stream 28: flow + temperature checked only - see remarks OK dates: available 2028-11-15, needed 2028-12-01: float 16 d

IF-04 Compressor wash water

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / utility
Medium / location	demin water / wash skid inlet
Size / connection	2 in (DN50) CL150 / ASME B16.5 CL150 RF flange
Scope break	IEC wash skid inlet flange; EPC demin water line, counter-flange, gasket and bolts.
Istanbul EPC side	EPC: demin water
Imaginary Electric side	IEC: wash skid inlet
Isolation	EPC manual valve; IEC skid inlet valve.
Protection	EPC demin pumps shut-off <= 8 barg.
Design	10.0 barg / 50 degC
Operating envelope	3.0 to 6.0 barg; 10 to 40 degC
Flow	normal 5.00 / max 10.00 m3/h (density 1000.0 kg/m3)
Quality / specification	requirement:ER-09.03, source:SRC-EPCE-0010
Conditions (text)	5 m3 per wash (off-line), tank filling at up to 10 m3/h.
Dates	available by 2028-09-15; needed by 2028-12-15
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0008
Checks (engine)	OK completeness: utility: complete OK dates: available 2028-09-15, needed 2028-12-15: float 91 d

IF-05 HP steam

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / process
Medium / location	steam / HRSG HP SH outlet stop valve, weld end
Size / connection	12 in P91 (SA-335 P91) / butt-weld end ASME B16.25, IEC pup piece 300 mm
Scope break	IEC supplies the HP SH outlet stop valve with a 300 mm pup piece (matching material and wall); the field weld, its PWHT and 100 % RT/UT are by EPC, witnessed by IEC.
Istanbul EPC side	EPC: main steam piping (EPC)
Imaginary Electric side	IEC: HRSG HP SH outlet stop valve, weld end
Isolation	IEC HP SH outlet motor-operated stop valve and non-return valve.
Protection	IEC HRSG HP SH safety valves set <= 196 bar(a), electromatic relief 194 bar(a).
Design	209.0 barg / 610 degC
Operating envelope	59.0 to 192.0 barg; 560 to 600 degC
Flow	normal 100.16 / max 105.00 kg/s
HMB envelope (all cases)	stream 6 (full check): 50.31-103.34 kg/s, 92.706-191.071 barg, 577.4-600.0 degC
Quality / specification	requirement:ER-09.03, source:SRC-EPCE-0010
Conditions (text)	Sliding pressure 60-193 bar(a); attemperation to 600 degC.
Dates	available by 2028-09-30; needed by 2028-10-16
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0005, source:SRC-IECE-0008
Checks (engine)	OK completeness: process: complete OK HMB flow: stream 6 max 103.34 kg/s (WIN-NG-100) <= capacity 105.00 kg/s (margin +1.6 %), 9 cases OK HMB pressure: stream 6: 92.706 - 191.071 barg within 59.0 - 192.0 barg (design 209.0) OK HMB temperature: stream 6: 577.4 - 600.0 degC within 560.0 - 600.0 degC (design 610.0) OK dates: available 2028-09-30, needed 2028-10-16: float 16 d

IF-06 HP steam

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / process
Medium / location	steam / ST main stop valve inlet, weld end
Size / connection	12 in P91 (SA-335 P91) / butt-weld end ASME B16.25, IEC pup piece 300 mm
Scope break	IEC supplies the ST main stop valve with pup piece; field weld, PWHT and NDT by EPC.
Istanbul EPC side	EPC: main steam piping (EPC)
Imaginary Electric side	IEC: ST main stop valve inlet, weld end
Isolation	IEC main stop valve (ST trip).
Protection	Upstream HRSG HP SH safety valves (IF-05) <= 196 bar(a); MSV casing design 215 bar(a) / 610 degC (TQ-IEC-021 item 1).
Design	214.0 barg / 610 degC
Operating envelope	52.0 to 186.0 barg; 560 to 600 degC
Flow	normal 100.16 / max 105.00 kg/s
HMB envelope (all cases)	stream 7 (full check): 50.31-103.34 kg/s, 89.936-185.369 barg, 575.9-598.5 degC
Fed from	IF-05 (chain dp 6.0 bar)
Quality / specification	requirement:ER-09.03, source:SRC-EPCE-0010
Conditions (text)	ST rated inlet 180 bar(a) / 598 degC per IEC 60045-1; main steam line pressure drop <= 6.0 bar at 105 kg/s.
Dates	available by 2028-09-30; needed by 2028-10-16
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0005, source:SRC-IECE-0008

Checks (engine)	OK completeness: process: complete OK HMB flow: stream 7 max 103.34 kg/s (WIN-NG-100) <= capacity 105.00 kg/s (margin +1.6 %), 9 cases OK HMB pressure: stream 7: 89.936 - 185.369 barg within 52.0 - 186.0 barg (design 214.0) OK HMB temperature: stream 7: 575.9 - 598.5 degC within 560.0 - 600.0 degC (design 610.0) OK chain pressure: IF-05 min 59.0 barg - 6.0 bar = 53.00 barg >= required 52.0 barg OK chain flow: capacity here 105.00 kg/s <= IF-05 105.00 kg/s OK dates: available 2028-09-30, needed 2028-10-16: float 16 d
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IF-07 Cold reheat

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / process
Medium / location	steam / HP turbine exhaust, weld end
Size / connection	24 in SA-335 P22 / butt-weld end, IEC pup piece
Scope break	IEC supplies the HP turbine exhaust nozzle with pup piece and the CRH non-return valve; field weld by EPC.
Istanbul EPC side	EPC: cold reheat piping (EPC)
Imaginary Electric side	IEC: HP turbine exhaust, weld end
Isolation	IEC CRH non-return valve (power assisted).
Protection	Reheater safety valves on the HRSG (IEC).
Design	51.0 barg / 420 degC
Operating envelope	18.0 to 41.0 barg; 340 to 395 degC
Flow	normal 100.16 / max 105.00 kg/s
HMB envelope (all cases)	stream 8 (full check): 50.31-103.34 kg/s, 19.633-40.476 barg, 357.1-387.6 degC
Quality / specification	requirement:ER-09.03, source:SRC-EPCE-0010
Dates	available by 2028-09-30; needed by 2028-10-16
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0005, source:SRC-IECE-0008
Checks (engine)	OK completeness: process: complete OK HMB flow: stream 8 max 103.34 kg/s (WIN-NG-100) <= capacity 105.00 kg/s (margin +1.6 %), 9 cases OK HMB pressure: stream 8: 19.633 - 40.476 barg within 18.0 - 41.0 barg (design 51.0) OK HMB temperature: stream 8: 357.1 - 387.6 degC within 340.0 - 395.0 degC (design 420.0) OK dates: available 2028-09-30, needed 2028-10-16: float 16 d

IF-08 Cold reheat

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / process
Medium / location	steam / HRSG reheater inlet, weld end
Size / connection	24 in SA-335 P22 / butt-weld end, IEC pup piece
Scope break	IEC supplies the HRSG reheater inlet with pup piece and the CRH safety valves; field weld by EPC.
Istanbul EPC side	EPC: cold reheat piping (EPC)
Imaginary Electric side	IEC: HRSG reheater inlet, weld end
Isolation	None at the HRSG inlet (CRH non-return valve at IF-07).
Protection	IEC cold reheat safety valves set 50 bar(a).
Design	51.0 barg / 420 degC
Operating envelope	17.0 to 41.0 barg; 340 to 395 degC
Flow	normal 100.16 / max 105.00 kg/s
HMB envelope (all cases)	stream 9 (full check): 50.31-103.34 kg/s, 19.289-39.754 barg, 356.1-386.6 degC
Fed from	IF-07 (chain dp 1.0 bar)
Quality / specification	requirement:ER-09.03, source:SRC-EPCE-0010
Dates	available by 2028-09-30; needed by 2028-10-16
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0005, source:SRC-IECE-0008
Checks (engine)	OK completeness: process: complete OK HMB flow: stream 9 max 103.34 kg/s (WIN-NG-100) <= capacity 105.00 kg/s (margin +1.6 %), 9 cases OK HMB pressure: stream 9: 19.289 - 39.754 barg within 17.0 - 41.0 barg (design 51.0) OK HMB temperature: stream 9: 356.1 - 386.6 degC within 340.0 - 395.0 degC (design 420.0) OK chain pressure: IF-07 min 18.0 barg - 1.0 bar = 17.00 barg >= required 17.0 barg OK chain flow: capacity here 105.00 kg/s <= IF-07 105.00 kg/s OK dates: available 2028-09-30, needed 2028-10-16: float 16 d

IF-09 Hot reheat

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / process
Medium / location	steam / HRSG reheater outlet stop valve, weld end
Size / connection	28 in P91 (SA-335 P91) / butt-weld end, IEC pup piece
Scope break	IEC supplies the HRSG reheater outlet stop valve with pup piece; field weld, PWHT, NDT by EPC.
Istanbul EPC side	EPC: hot reheat piping (EPC)
Imaginary Electric side	IEC: HRSG reheater outlet stop valve, weld end
Isolation	IEC HRH outlet stop valve.
Protection	IEC hot reheat safety valves set <= 48 bar(a).

Design	49.0 barg / 610 degC
Operating envelope	17.0 to 38.5 barg; 550 to 600 degC
Flow	normal 117.50 / max 125.00 kg/s
HMB envelope (all cases)	stream 11 (full check): 60.92-121.56 kg/s, 18.294-37.783 barg, 566.0-600.0 degC
Quality / specification	requirement:ER-09.03, source:SRC-EPCE-0010
Dates	available by 2028-09-30; needed by 2028-10-16
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0005, source:SRC-IECE-0008
Checks (engine)	OK completeness: process: complete OK HMB flow: stream 11 max 121.56 kg/s (WIN-NG-100) <= capacity 125.00 kg/s (margin +2.8 %), 9 cases OK HMB pressure: stream 11: 18.294 - 37.783 barg within 17.0 - 38.5 barg (design 49.0) OK HMB temperature: stream 11: 566.0 - 600.0 degC within 550.0 - 600.0 degC (design 610.0) OK dates: available 2028-09-30, needed 2028-10-16: float 16 d

IF-10 Hot reheat

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / process
Medium / location	steam / ST IP stop-control valve inlet
Size / connection	28 in P91 (SA-335 P91) / butt-weld end, IEC pup piece
Scope break	IEC supplies the ST IP stop/control valve with pup piece; field weld, PWHT, NDT by EPC.
Istanbul EPC side	EPC: hot reheat piping (EPC)
Imaginary Electric side	IEC: ST IP stop-control valve inlet
Isolation	IEC IP stop valve (ST trip).
Protection	Upstream HRH safety valves (IF-09) <= 48 bar(a); IP valve casing design 52 bar(a) / 610 degC (TQ-IEC-021 item 1).
Design	51.0 barg / 610 degC
Operating envelope	15.5 to 38.5 barg; 550 to 600 degC
Flow	normal 117.50 / max 125.00 kg/s
HMB envelope (all cases)	stream 12 (full check): 60.92-121.56 kg/s, 17.561-36.331 barg, 564.5-598.5 degC
Fed from	IF-09 (chain dp 1.5 bar)
Quality / specification	requirement:ER-09.03, source:SRC-EPCE-0010
Dates	available by 2028-09-30; needed by 2028-10-16
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0005, source:SRC-IECE-0008
Checks (engine)	OK completeness: process: complete OK HMB flow: stream 12 max 121.56 kg/s (WIN-NG-100) <= capacity 125.00 kg/s (margin +2.8 %), 9 cases OK HMB pressure: stream 12: 17.561 - 36.331 barg within 15.5 - 38.5 barg (design 51.0) OK HMB temperature: stream 12: 564.5 - 598.5 degC within 550.0 - 600.0 degC (design 610.0) OK chain pressure: IF-09 min 17.0 barg - 1.5 bar = 15.50 barg >= required 15.5 barg OK chain flow: capacity here 125.00 kg/s <= IF-09 125.00 kg/s OK dates: available 2028-09-30, needed 2028-10-16: float 16 d

IF-11 LP steam

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / process
Medium / location	steam / HRSG LP SH outlet stop valve
Size / connection	20 in SA-106 C / butt-weld end
Scope break	IEC supplies the LP SH outlet stop valve with pup piece; field weld by EPC.
Istanbul EPC side	EPC: LP steam piping (EPC)
Imaginary Electric side	IEC: HRSG LP SH outlet stop valve
Isolation	IEC LP SH outlet stop valve and non-return valve.
Protection	IEC LP SH safety valve set 8.0 bar(a).
Design	9.0 barg / 330 degC
Operating envelope	1.5 to 5.0 barg; 210 to 260 degC
Flow	normal 11.39 / max 16.50 kg/s
HMB envelope (all cases)	stream 13 (full check): 6.36-15.27 kg/s, 1.783-4.586 barg, 218.7-251.8 degC
Quality / specification	requirement:ER-09.03, source:SRC-EPCE-0010
Dates	available by 2028-09-30; needed by 2028-10-16
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0005, source:SRC-IECE-0008
Checks (engine)	OK completeness: process: complete OK HMB flow: stream 13 max 15.27 kg/s (SRC-LDO-100) <= capacity 16.50 kg/s (margin +7.5 %), 9 cases OK HMB pressure: stream 13: 1.783 - 4.586 barg within 1.5 - 5.0 barg (design 9.0) OK HMB temperature: stream 13: 218.7 - 251.8 degC within 210.0 - 260.0 degC (design 330.0) OK dates: available 2028-09-30, needed 2028-10-16: float 16 d

IF-12 LP steam

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / process
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Medium / location	steam / ST LP admission valve inlet
Size / connection	20 in SA-106 C / butt-weld end
Scope break	IEC supplies the ST LP admission valve with pup piece; field weld by EPC.
Istanbul EPC side	EPC: LP steam piping (EPC)
Imaginary Electric side	IEC: ST LP admission valve inlet
Isolation	IEC LP admission stop valve.
Protection	Upstream LP SH safety valve (IF-11) 8.0 bar(a); admission valve casing design 10 bar(a) / 330 degC (TQ-IEC-021 item 1).
Design	9.0 barg / 330 degC
Operating envelope	1.1 to 5.0 barg; 210 to 260 degC
Flow	normal 11.39 / max 16.50 kg/s
HMB envelope (all cases)	stream 14 (full check): 6.36-15.27 kg/s, 1.589-4.219 barg, 217.7-250.8 degC
Fed from	IF-11 (chain dp 0.4 bar)
Quality / specification	requirement:ER-09.03, source:SRC-EPCE-0010
Dates	available by 2028-09-30; needed by 2028-10-16
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0005, source:SRC-IECE-0008
Checks (engine)	OK completeness: process: complete OK HMB flow: stream 14 max 15.27 kg/s (SRC-LDO-100) <= capacity 16.50 kg/s (margin +7.5 %), 9 cases OK HMB pressure: stream 14: 1.589 - 4.219 barg within 1.1 - 5.0 barg (design 9.0) OK HMB temperature: stream 14: 217.7 - 250.8 degC within 210.0 - 260.0 degC (design 330.0) OK chain pressure: IF-11 min 1.5 barg - 0.4 bar = 1.10 barg >= required 1.1 barg OK chain flow: capacity here 16.50 kg/s <= IF-11 16.50 kg/s OK dates: available 2028-09-30, needed 2028-10-16: float 16 d

IF-13 HP / IP / LP bypass valves

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / process
Medium / location	steam / water / valves supplied loose by IEC (incl. desuperheating)
Size / connection	HP 12 in / IP 28 in / LP 20 in (per valve datasheets) / butt-weld ends; spray water flanges CL2500 (HP) / CL600 (IP, LP)
Scope break	Loose supply: IEC supplies HP, IP and LP bypass valves with desuperheaters (free issue to site); EPC installs them in its piping, supplies the spray water lines from the BFP discharge/interstage and CEP discharge and all welds.
Istanbul EPC side	EPC: installation in EPC piping, spray water from BFP/CEP (EPC)
Imaginary Electric side	IEC: valves supplied loose by IEC (incl. desuperheating)
Isolation	EPC isolation on the spray water lines; bypass valves close on loss of spray water (IEC logic).
Protection	HP bypass downstream design 52 bar(a) (CRH); IP/LP bypass downstream 16 bar(a) (IF-14).
Design	214.0 barg / 610 degC
Operating envelope	1.0 to 192.0 barg; 150 to 600 degC
Flow	normal - / max 125.00 kg/s
Quality / specification	requirement:ER-09.03, source:SRC-EPCE-0010
Conditions (text)	Inlet side most onerous (HP bypass inlet); delivery of the loose valves by 2028-02-01 for piping erection.
Dates	available by 2028-02-01; needed by 2028-03-01
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0008
Checks (engine)	OK completeness: process: complete OK dates: available 2028-02-01, needed 2028-03-01: float 29 d

IF-14 Bypass steam to condenser

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / process
Medium / location	steam / condenser neck dump devices (IEC)
Size / connection	2 x 36 in + 1 x 28 in / butt-weld end at the condenser neck dump devices
Scope break	IEC supplies the dump devices in the condenser neck with weld ends; EPC supplies the IP/LP bypass downstream piping and the welds.
Istanbul EPC side	EPC: IP/LP bypass downstream piping (EPC)
Imaginary Electric side	IEC: condenser neck dump devices (IEC)
Isolation	Bypass valves (IF-13).
Protection	Condenser rupture discs (IEC) and bypass trip on high condenser pressure.
Design	15.0 barg / 250 degC
Operating envelope	-1.0 to 12.0 barg; 40 to 250 degC
Flow	normal - / max 141.50 kg/s
Quality / specification	requirement:ER-09.03, source:SRC-EPCE-0010
Conditions (text)	Start-up and trip only; 100 % HRH + LP flow.
Dates	available by 2028-05-01; needed by 2028-06-01
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0008
Checks (engine)	OK completeness: process: complete OK dates: available 2028-05-01, needed 2028-06-01: float 31 d

IF-15 Condensate from hotwell

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / process
Medium / location	condensate / hotwell outlet nozzle
Size / connection	20 in (DN500) CL150 / ASME B16.5 CL150 RF flange, vacuum-tight
Scope break	IEC hotwell outlet nozzle flange; EPC counter-flange, spiral-wound gasket, bolts and CEP suction piping.
Istanbul EPC side	EPC: CEP suction piping (EPC)
Imaginary Electric side	IEC: hotwell outlet nozzle
Isolation	EPC motor-operated suction valve per CEP.
Protection	Full vacuum design both sides.
Design	2.5 barg / 80 degC
Operating envelope	-0.995 to -0.900 barg; 15 to 40 degC
Flow	normal 128.00 / max 140.00 kg/s
HMB envelope (all cases)	stream 17 (full check): 67.28-133.94 kg/s, -0.993--0.952 barg, 17.5-36.5 degC
Quality / specification	requirement:ER-09.03, source:SRC-EPCE-0010
Conditions (text)	Full vacuum to 3.5 bar(a).
Dates	available by 2028-09-01; needed by 2028-09-15
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0005, source:SRC-IECE-0008
Checks (engine)	OK completeness: process: complete OK HMB flow: stream 17 max 133.94 kg/s (WIN-NG-100) <= capacity 140.00 kg/s (margin +4.3 %), 9 cases OK HMB pressure: stream 17: -0.993 - -0.952 barg within -0.995 - -0.9 barg (design 2.5) OK HMB temperature: stream 17: 17.5 - 36.5 degC within 15.0 - 40.0 degC (design 80.0) OK dates: available 2028-09-01, needed 2028-09-15: float 14 d

IF-16 Condensate to CPH

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / process
Medium / location	condensate / HRSG CPH inlet
Size / connection	12 in (DN300) CL300 / ASME B16.5 CL300 RF flange
Scope break	IEC HRSG condensate preheater inlet flange; EPC condensate line, counter-flange, gasket, bolts.
Istanbul EPC side	EPC: condensate line (EPC)
Imaginary Electric side	IEC: HRSG CPH inlet
Isolation	EPC motor-operated isolation valve.
Protection	EPC CEP shut-off head <= 22 bar(a) at maximum speed (< 25 bar(a) design), verified at pump FAT.
Design	24.0 barg / 180 degC
Operating envelope	12.0 to 21.0 barg; 15 to 45 degC
Flow	normal 141.37 / max 150.00 kg/s
HMB envelope (all cases)	stream 20 (full check): 73.89-147.33 kg/s, 14.487-14.487 barg, 23.0-40.0 degC
Quality / specification	requirement:ER-09.03, source:SRC-EPCE-0010
Dates	available by 2028-09-01; needed by 2028-09-15
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0005, source:SRC-IECE-0008
Checks (engine)	OK completeness: process: complete OK HMB flow: stream 20 max 147.33 kg/s (WIN-NG-100) <= capacity 150.00 kg/s (margin +1.8 %), 9 cases OK HMB pressure: stream 20: 14.487 - 14.487 barg within 12.0 - 21.0 barg (design 24.0) OK HMB temperature: stream 20: 23.0 - 40.0 degC within 15.0 - 45.0 degC (design 180.0) OK dates: available 2028-09-01, needed 2028-09-15: float 14 d

IF-17 BFP suction

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / process
Medium / location	feedwater / HRSG LP drum downcomer outlet nozzle
Size / connection	16 in (DN400) CL150 / ASME B16.5 CL150 RF flange
Scope break	IEC LP drum downcomer outlet nozzle flange; EPC BFP suction piping, counter-flange, gasket, bolts.
Istanbul EPC side	EPC: BFP suction piping (EPC)
Imaginary Electric side	IEC: HRSG LP drum downcomer outlet nozzle
Isolation	EPC suction isolation valve per BFP.
Protection	IEC LP drum safety valves.
Design	9.0 barg / 185 degC
Operating envelope	1.5 to 6.0 barg; 130 to 165 degC
Flow	normal 129.98 / max 140.00 kg/s
HMB envelope (all cases)	stream 21 (full check): 67.54-134.94 kg/s, 2.059-5.294 barg, 134.3-160.7 degC
Quality / specification	requirement:ER-09.03, source:SRC-EPCE-0010
Conditions (text)	LP drum normal water level EL +31.5 m; the EPC confirms BFP NPSH with the drum level and suction losses.
Dates	available by 2028-09-01; needed by 2028-09-15
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0005, source:SRC-IECE-0008

Checks (engine)	OK completeness: process: complete OK HMB flow: stream 21 max 134.94 kg/s (WIN-NG-100) <= capacity 140.00 kg/s (margin +3.6 %), 9 cases OK HMB pressure: stream 21: 2.059 - 5.294 barg within 1.5 - 6.0 barg (design 9.0) OK HMB temperature: stream 21: 134.3 - 160.7 degC within 130.0 - 165.0 degC (design 185.0) OK dates: available 2028-09-01, needed 2028-09-15: float 14 d
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IF-18 HP feedwater

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / process
Medium / location	feedwater / HRSG HP ECO1 inlet
Size / connection	10 in (DN250) CL2500 / butt-weld end (SA-106 C / SA-335 P11)
Scope break	IEC HRSG HP economiser 1 inlet with pup piece; field weld by EPC.
Istanbul EPC side	EPC: HP feedwater piping (EPC)
Imaginary Electric side	IEC: HRSG HP ECO1 inlet
Isolation	EPC HP feedwater motor-operated isolation valve and non-return valve.
Protection	EPC BFP shut-off head at maximum speed <= 234 barg (VFD speed limit), verified at pump FAT; minimum-flow recirculation valve.
Design	234.0 barg / 200 degC
Operating envelope	95.0 to 215.0 barg; 130 to 170 degC
Flow	normal 100.16 / max 110.00 kg/s
HMB envelope (all cases)	stream 22 (full check): 50.31-103.34 kg/s, 100.266-214.270 barg, 135.9-163.9 degC
Quality / specification	requirement:ER-09.03, source:SRC-EPCE-0010
Dates	available by 2028-09-01; needed by 2028-09-15
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0005, source:SRC-IECE-0008
Checks (engine)	OK completeness: process: complete OK HMB flow: stream 22 max 103.34 kg/s (WIN-NG-100) <= capacity 110.00 kg/s (margin +6.1 %), 9 cases OK HMB pressure: stream 22: 100.266 - 214.270 barg within 95.0 - 215.0 barg (design 234.0) OK HMB temperature: stream 22: 135.9 - 163.9 degC within 130.0 - 170.0 degC (design 200.0) OK dates: available 2028-09-01, needed 2028-09-15: float 14 d

IF-19 IP feedwater

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / process
Medium / location	feedwater / HRSG IP ECO inlet
Size / connection	6 in (DN150) CL600 / butt-weld end
Scope break	IEC HRSG IP economiser inlet with pup piece; field weld by EPC.
Istanbul EPC side	EPC: IP feedwater piping (EPC)
Imaginary Electric side	IEC: HRSG IP ECO inlet
Isolation	EPC IP feedwater isolation and non-return valve.
Protection	EPC BFP interstage pressure at shut-off <= 58 bar(a), verified at FAT.
Design	59.0 barg / 200 degC
Operating envelope	20.0 to 48.0 barg; 130 to 165 degC
Flow	normal 29.82 / max 33.00 kg/s
HMB envelope (all cases)	stream 23 (full check): 17.23-31.59 kg/s, 22.358-46.927 barg, 134.6-161.4 degC
Quality / specification	requirement:ER-09.03, source:SRC-EPCE-0010
Dates	available by 2028-09-01; needed by 2028-09-15
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0005, source:SRC-IECE-0008
Checks (engine)	OK completeness: process: complete OK HMB flow: stream 23 max 31.59 kg/s (WIN-NG-100) <= capacity 33.00 kg/s (margin +4.3 %), 9 cases OK HMB pressure: stream 23: 22.358 - 46.927 barg within 20.0 - 48.0 barg (design 59.0) OK HMB temperature: stream 23: 134.6 - 161.4 degC within 130.0 - 165.0 degC (design 200.0) OK dates: available 2028-09-01, needed 2028-09-15: float 14 d

IF-20 FGH water return

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / process
Medium / location	feedwater / performance gas heater water outlet
Size / connection	4 in (DN100) CL600 / ASME B16.5 CL600 RF flange
Scope break	IEC performance gas heater water outlet flange; EPC return line to the condensate system.
Istanbul EPC side	EPC: return line to condensate (EPC)
Imaginary Electric side	IEC: performance gas heater water outlet
Isolation	EPC isolation valve; IEC FGH outlet control valve.
Protection	EPC return line designed 60 bar(a) / 260 degC up to and including its pressure-reducing valve at the condensate connection; condensate system relief valve downstream (EPC).
Design	59.0 barg / 260 degC
Operating envelope	18.0 to 46.0 barg; 60 to 90 degC
Flow	normal 12.48 / max 14.50 kg/s

HMB envelope (all cases)	stream 19 (full check): 6.62-13.40 kg/s, 19.358-43.927 barg, 75.0-75.0 degC
Quality / specification	requirement:ER-09.03, source:SRC-EPCE-0010
Dates	available by 2028-09-01; needed by 2028-09-15
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0005, source:SRC-IECE-0008
Checks (engine)	OK completeness: process: complete OK HMB flow: stream 19 max 13.40 kg/s (MIN-NG-100) <= capacity 14.50 kg/s (margin +7.6 %), 8 cases OK HMB pressure: stream 19: 19.358 - 43.927 barg within 18.0 - 46.0 barg (design 59.0) OK HMB temperature: stream 19: 75.0 - 75.0 degC within 60.0 - 90.0 degC (design 260.0) OK dates: available 2028-09-01, needed 2028-09-15: float 14 d

IF-21 HRSG blowdown

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / process
Medium / location	water / continuous / intermittent blowdown valves outlet
Size / connection	2 in / 3 in / socket-weld / butt-weld ends
Scope break	IEC supplies the continuous and intermittent blowdown valves (two in series); scope break at the outlet weld of the second valve; EPC blowdown piping, flash tanks and silencers.
Istanbul EPC side	EPC: blowdown tanks (EPC)
Imaginary Electric side	IEC: continuous / intermittent blowdown valves outlet
Isolation	IEC two valves in series (ASME I PG-58.3.6).
Protection	EPC blowdown piping designed for the drum design pressure up to the flash tank inlet.
Design	209.0 barg / 370 degC
Operating envelope	0.000 to 5.0 barg; 100 to 370 degC
Flow	normal 1.00 / max 5.00 kg/s
Quality / specification	requirement:ER-09.03, source:SRC-EPCE-0010
Conditions (text)	Continuous blowdown up to 1 % of HP flow; intermittent 5 kg/s for 30 s.
Dates	available by 2028-09-01; needed by 2028-09-15
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0008
Checks (engine)	OK completeness: process: complete OK dates: available 2028-09-01, needed 2028-09-15: float 14 d

IF-22 HRSG drains and vents

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / process
Medium / location	water/steam / drain/vent valves outlet (IEC)
Size / connection	various (1 in - 3 in) / socket-weld / butt-weld ends
Scope break	Scope break at the outlet weld of the IEC drain/vent valve (second isolation); EPC drain collection, atmospheric flash tanks and vent silencers.
Istanbul EPC side	EPC: drain collection, flash tanks, silencers (EPC)
Imaginary Electric side	IEC: drain/vent valves outlet (IEC)
Isolation	IEC two valves in series on all pressure part drains.
Protection	EPC drain headers designed for the highest upstream design up to the first flash tank.
Design	209.0 barg / 610 degC
Operating envelope	0.000 to 1.0 barg; 40 to 610 degC
Flow	normal - / max 10.00 kg/s
Quality / specification	requirement:ER-09.03, source:SRC-EPCE-0010
Conditions (text)	Start-up drains.
Dates	available by 2028-09-01; needed by 2028-09-15
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0008
Checks (engine)	OK completeness: process: complete OK dates: available 2028-09-01, needed 2028-09-15: float 14 d

IF-23 Chemical dosing

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / utility
Medium / location	chemicals / dosing nozzles on drums/condensate
Size / connection	1 in / socket-weld, IEC root valve
Scope break	IEC dosing nozzles with root valves on the drums; EPC dosing skids and lines.
Istanbul EPC side	EPC: dosing skids and lines (EPC)
Imaginary Electric side	IEC: dosing nozzles on drums/condensate
Isolation	IEC root valve + EPC isolation and check valve.
Protection	EPC dosing pump relief set below the line design.
Design	234.0 barg / 100 degC
Operating envelope	3.0 to 215.0 barg; 5 to 40 degC
Flow	normal - / max 0.10 m3/h (density 1000.0 kg/m3)
Quality / specification	source:SRC-EPCE-0010
Conditions (text)	Lay-up and optional drum dosing; AVT(O) ammonia dosing into the condensate by EPC.
Dates	available by 2028-09-01; needed by 2028-09-15

Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0008
Checks (engine)	OK completeness: utility: complete OK dates: available 2028-09-01, needed 2028-09-15: float 14 d

IF-24 Sampling

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / utility
Medium / location	water/steam / sample nozzles with root valves (IEC)
Size / connection	1/2 in / socket-weld, IEC root valve
Scope break	IEC sample nozzles with root valves; EPC sample lines, coolers and the SWAS.
Istanbul EPC side	EPC: sample lines, coolers, SWAS (EPC)
Imaginary Electric side	IEC: sample nozzles with root valves (IEC)
Isolation	IEC root valve; EPC isolation at the SWAS.
Protection	EPC sample line design = upstream design.
Design	234.0 barg / 610 degC
Operating envelope	-1.0 to 215.0 barg; 20 to 600 degC
Flow	normal - / max 0.50 kg/s
Quality / specification	source:SRC-EPCE-0010
Conditions (text)	Sample points per IEC-ALP-REQ-001 section 4.
Dates	available by 2028-09-01; needed by 2028-09-15
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0008
Checks (engine)	OK completeness: utility: complete OK dates: available 2028-09-01, needed 2028-09-15: float 14 d

IF-25 Aqueous ammonia

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / utility
Medium / location	NH4OH 24.5 % / ammonia injection skid inlet (IEC)
Size / connection	1 in / ASME B16.5 CL150 RF flange (316L)
Scope break	IEC ammonia injection skid inlet flange; EPC storage, unloading and forwarding pumps.
Istanbul EPC side	EPC: storage, unloading, forwarding pumps (EPC)
Imaginary Electric side	IEC: ammonia injection skid inlet (IEC)
Isolation	EPC ESD valve (fail-closed) and manual valve.
Protection	EPC forwarding pump relief 9 barg.
Design	10.0 barg / 40 degC
Operating envelope	3.0 to 7.0 barg; 5 to 40 degC
Flow	normal 0.12 / max 0.50 t/h
Quality / specification	requirement:ER-06.03, source:SRC-EPCE-0010
Conditions (text)	Aqueous ammonia 24.5 %: 125 kg/h on gas, 480 kg/h on LDO (TQ-IEC-018).
Dates	available by 2028-11-01; needed by 2028-11-15
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0008
Checks (engine)	OK completeness: utility: complete OK dates: available 2028-11-01, needed 2028-11-15: float 14 d

IF-26 Flue gas to stack

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / process
Medium / location	flue gas / HRSG outlet transition flange
Size / connection	D 9.2 m / bolted flange with EPC fabric expansion joint
Scope break	IEC HRSG outlet transition duct flange (drilled); EPC fabric expansion joint, stack, silencer, CEMS and damper (if any).
Istanbul EPC side	EPC: stack (EPC), CEMS (EPC)
Imaginary Electric side	IEC: HRSG outlet transition flange
Isolation	None (no stack damper).
Protection	HRSG casing design +/- 50 mbar.
Design	0.050 barg / 200 degC
Operating envelope	0.000 to 0.030 barg; 60 to 120 degC
Flow	normal 796.70 / max 850.00 kg/s
HMB envelope (all cases)	stream 5 (full check): 441.95-832.46 kg/s, 0.001-0.001 barg, 68.2-89.7 degC
Quality / specification	requirement:ER-14.02
Dates	available by 2028-06-30; needed by 2028-09-15
Scope items	SC-010, SC-011
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0005, source:SRC-IECE-0008

Checks (engine)	OK completeness: process: complete OK HMB flow: stream 5 max 832.46 kg/s (WIN-NG-100) <= capacity 850.00 kg/s (margin +2.1 %), 9 cases OK HMB pressure: stream 5: 0.001 - 0.001 barg within 0.0 - 0.03 barg (design 0.05) OK HMB temperature: stream 5: 68.2 - 89.7 degC within 60.0 - 120.0 degC (design 200.0) OK dates: available 2028-06-30, needed 2028-09-15: float 77 d
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IF-27 Circulating water

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / process
Medium / location	seawater / condenser water box flanges (2 inlets, 2 outlets)
Size / connection	4 x DN1800 / EN 1092-1 PN6 flanges, rubber-lined water boxes
Scope break	IEC condenser water box inlet/outlet flanges; EPC CW piping (GRP), expansion joints, counter-flanges and bolts.
Istanbul EPC side	EPC: CW pipes (EPC)
Imaginary Electric side	IEC: condenser water box flanges (2 inlets, 2 outlets)
Isolation	EPC butterfly valves per inlet and outlet half (condenser half-side operation).
Protection	EPC CW pump shut-off head <= 3.5 barg at the condenser; surge analysis for pump trip <= 4 barg; vacuum breaker at the outlet high point.
Design	4.0 barg / 40 degC
Operating envelope	0.500 to 3.0 barg; 5 to 35 degC
Flow	normal 36,050.00 / max 38,000.00 m3/h (density 1025.0 kg/m3)
HMB envelope (all cases)	stream 25 (full check): 10196.80-10196.80 kg/s, 1.487-1.487 barg, 6.9-27.0 degC
Quality / specification	source:SRC-OWNR-0002, requirement:ER-02.04
Conditions (text)	Design flow 38,000 m3/h (DEC-EPCE-0002); condenser pressure drop 0.62 bar clean.
Dates	available by 2028-09-01; needed by 2028-09-15
Scope items	SC-012
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0005, source:SRC-IECE-0008
Checks (engine)	OK completeness: process: complete OK HMB flow: stream 25 max 10196.80 kg/s (WIN-NG-100) <= capacity 10819.44 kg/s (margin +5.8 %), 9 cases OK HMB pressure: stream 25: 1.487 - 1.487 barg within 0.5 - 3.0 barg (design 4.0) OK HMB temperature: stream 25: 6.9 - 27.0 degC within 5.0 - 35.0 degC (design 40.0) OK dates: available 2028-09-01, needed 2028-09-15: float 14 d

IF-28 Closed cooling water

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / utility
Medium / location	demin water + inhibitor / IEC coolers inlet/outlet flanges
Size / connection	various (DN50 - DN400) / ASME B16.5 CL150 RF flanges
Scope break	IEC cooler inlet/outlet flanges; EPC CCW system.
Istanbul EPC side	EPC: CCW system (EPC)
Imaginary Electric side	IEC: IEC coolers inlet/outlet flanges
Isolation	EPC valves per consumer.
Protection	EPC CCW pump shut-off <= 8 barg.
Design	10.0 barg / 60 degC
Operating envelope	3.0 to 6.0 barg; 20 to 36 degC
Flow	normal 1,055.00 / max 1,160.00 m3/h (density 1000.0 kg/m3)
Quality / specification	source:SRC-EPCE-0010
Conditions (text)	9,400 kW; supply <= 36 degC (ST TEWAC) / 38 degC (others).
Dates	available by 2028-08-09; needed by 2028-08-16
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0008
Checks (engine)	OK completeness: utility: complete INFO dates: available 2028-08-09, needed 2028-08-16: float 7 d (tight)

IF-29 Instrument and service air

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / utility
Medium / location	air / IEC skids
Size / connection	1 in - 2 in / ASME B16.5 CL150 RF flanges
Scope break	IEC skid inlet flanges; EPC compressed air system.
Istanbul EPC side	EPC: compressed air system (EPC)
Imaginary Electric side	IEC: IEC skids
Isolation	EPC valve per skid.
Protection	EPC air receivers relief 10 barg.
Design	10.0 barg / 50 degC
Operating envelope	6.0 to 8.0 barg; 5 to 45 degC
Flow	normal 350.00 / max 500.00 Nm3/h (density 1.293 kg/m3)
Quality / specification	source:SRC-EPCE-0010
Conditions (text)	Instrument air dew point -40 degC at 7 barg (ISO 8573-1 class 1.2.1).

Dates	available by 2028-08-09; needed by 2028-08-16
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0008
Checks (engine)	OK completeness: utility: complete INFO dates: available 2028-08-09, needed 2028-08-16: float 7 d (tight)

IF-30 Nitrogen

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / utility
Medium / location	N2 / HRSG lay-up / generator purge connections
Size / connection	1 in / ASME B16.5 CL150 RF flange
Scope break	IEC lay-up / purge connections; EPC nitrogen supply.
Istanbul EPC side	EPC: N2 supply (EPC)
Imaginary Electric side	IEC: HRSG lay-up / generator purge connections
Isolation	EPC valve + IEC valve.
Protection	EPC pressure regulator + relief 9 barg.
Design	10.0 barg / 50 degC
Operating envelope	5.0 to 8.0 barg; 5 to 40 degC
Flow	normal - / max 1,000.00 Nm3/h (density 1.25 kg/m3)
Quality / specification	source:SRC-EPCE-0010
Conditions (text)	99.9 % N2; 3,000 Nm3 per lay-up event.
Dates	available by 2028-09-01; needed by 2028-09-15
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0008
Checks (engine)	OK completeness: utility: complete OK dates: available 2028-09-01, needed 2028-09-15: float 14 d

IF-31 Hydrogen / CO2

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / utility
Medium / location	H2 / CO2 / GT generator gas control panel inlet
Size / connection	1/2 in / compression fitting (316L)
Scope break	IEC generator gas control panel inlet; EPC bottle racks, regulators and piping.
Istanbul EPC side	EPC: bottle racks and piping (EPC)
Imaginary Electric side	IEC: GT generator gas control panel inlet
Isolation	EPC valves at the racks.
Protection	EPC regulator + relief 14 barg.
Design	16.0 barg / 50 degC
Operating envelope	8.0 to 12.0 barg; 5 to 40 degC
Flow	normal - / max 60.00 Nm3/h
Quality / specification	source:SRC-EPCE-0010
Conditions (text)	H2 99.9 %, 12 Nm3/day; CO2 150 Nm3 per purge.
Dates	available by 2028-08-25; needed by 2028-09-01
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0008
Checks (engine)	OK completeness: utility: complete INFO dates: available 2028-08-25, needed 2028-09-01: float 7 d (tight)

IF-32 Auxiliary steam

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / utility
Medium / location	steam / gland steam header inlet (IEC)
Size / connection	4 in (DN100) CL300 / butt-weld end
Scope break	IEC gland steam header inlet; EPC auxiliary steam line from the auxiliary boiler.
Istanbul EPC side	EPC: auxiliary boiler steam (EPC)
Imaginary Electric side	IEC: gland steam header inlet (IEC)
Isolation	EPC isolation valve.
Protection	EPC auxiliary steam header relief set 15 barg.
Design	15.0 barg / 320 degC
Operating envelope	8.0 to 9.5 barg; 250 to 300 degC
Flow	normal 4.50 / max 7.50 t/h
Quality / specification	requirement:ER-09.03, source:SRC-EPCE-0010
Conditions (text)	4.5 t/h gland sealing, 3 t/h HRSG sparging (optional).
Dates	available by 2028-09-30; needed by 2028-10-16
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0008
Checks (engine)	OK completeness: utility: complete OK dates: available 2028-09-30, needed 2028-10-16: float 16 d

IF-33 Fire protection

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / control
Medium / location	- / GT enclosure CO2 system (IEC)
Size / connection	- / hardwired volt-free contacts + Modbus TCP summary
Scope break	IEC GT enclosure CO2 fire protection panel terminals; EPC fire alarm panel (NFPA 72), cables, transformer deluge, hydrants.
Istanbul EPC side	EPC: fire detection/alarm, transformer deluge, hydrants (EPC)
Imaginary Electric side	IEC: GT enclosure CO2 system (IEC)
Isolation	-
Protection	-
Dates	available by 2028-08-17; needed by 2028-08-24
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0008
Checks (engine)	OK completeness: control: complete INFO dates: available 2028-08-17, needed 2028-08-24: float 7 d (tight)

IF-34 GT GSU HV

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / electrical
Medium / location	electrical / GT GSU HV bushings
Size / connection	560 MVA / GSU HV air bushings, EPC 420 kV connection to the GIS
Scope break	IEC GSU HV bushing terminals; EPC conductors/cable, clamps, surge arresters at the bushings and the GIS bay.
Istanbul EPC side	EPC: 380 kV connection to GIS (EPC)
Imaginary Electric side	IEC: GT GSU HV bushings
Isolation	EPC GIS generator transformer bay circuit breaker, disconnecter and earthing switch; IEC GCB on the LV side.
Protection	GSU differential (87T) and overall differential (87G-T) by IEC; GIS bay protection by EPC; trip matrix per IF-41.
Flow	normal - / max 560.00 MVA
Electrical	380.00 kV (342.0-420.0 kV); Isc 50.0 kA, rating 50.0 kA; pf 0.85
HMB power	gt_output
Fed from	TP-E1 (chain dp - bar)
Conditions (text)	OLTC 400 +/- 8 x 1.25 % (360-440 kV) with generator 21 kV +/- 5 % covers 342-420 kV.
Dates	available by 2028-07-29; needed by 2028-08-02
Scope items	SC-014
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0008
Checks (engine)	OK completeness: electrical: complete OK short circuit: prospective 50.0 kA <= rating 50.0 kA OK HMB power: max gt_output 470.0 MW (WIN-NG-100) / pf 0.85 = 552.9 MVA <= 560 MVA (margin +1.3 %) OK chain voltage: TP-E1 342.0-420.0 kV within 342.0-420.0 kV here OK chain short circuit: TP-E1 50.0 kA <= rating 50.0 kA here INFO dates: available 2028-07-29, needed 2028-08-02: float 4 d (tight)

IF-35 ST GSU HV

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / electrical
Medium / location	electrical / ST GSU HV bushings
Size / connection	270 MVA / GSU HV air bushings, EPC 420 kV connection to the GIS
Scope break	As IF-34 for the ST GSU.
Istanbul EPC side	EPC: 380 kV connection to GIS (EPC)
Imaginary Electric side	IEC: ST GSU HV bushings
Isolation	As IF-34.
Protection	As IF-34.
Flow	normal - / max 270.00 MVA
Electrical	380.00 kV (342.0-420.0 kV); Isc 50.0 kA, rating 50.0 kA; pf 0.85
HMB power	st_output
Fed from	TP-E1 (chain dp - bar)
Conditions (text)	OLTC 400 +/- 8 x 1.25 % with generator 15.75 kV +/- 5 %.
Dates	available by 2028-07-29; needed by 2028-08-02
Scope items	SC-014
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0008
Checks (engine)	OK completeness: electrical: complete OK short circuit: prospective 50.0 kA <= rating 50.0 kA OK HMB power: max st_output 207.9 MW (WIN-NG-100) / pf 0.85 = 244.6 MVA <= 270 MVA (margin +9.4 %) OK chain voltage: TP-E1 342.0-420.0 kV within 342.0-420.0 kV here OK chain short circuit: TP-E1 50.0 kA <= rating 50.0 kA here INFO dates: available 2028-07-29, needed 2028-08-02: float 4 d (tight)

IF-36 UAT supply

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / electrical
Medium / location	electrical / IPB tap-off flange (GT and ST IPB)
Size / connection	21 kV (GT) / 15.75 kV (ST) / IPB tap-off flange (IEC), EPC MV busduct
Scope break	IEC IPB tap-off flange between GCB and GSU; EPC UAT, UAT busduct and flexible links.
Istanbul EPC side	EPC: UAT and MV busduct (EPC)
Imaginary Electric side	IEC: IPB tap-off flange (GT and ST IPB)
Isolation	IEC tap-off disconnecter link; EPC UAT MV circuit breaker.
Protection	UAT differential and overcurrent by EPC, integrated in the IEC unit trip matrix.
Flow	normal - / max 31.50 MVA
Electrical	21.00 kV (19.9-22.1 kV); Isc 169.0 kA, rating 200.0 kA
Conditions (text)	Prospective 169 kA at the GT generator bus (96 kA via GSU + 73 kA generator); ST bus 125 kA. IPB tap-off rated 200 kA sym / 550 kA peak (TQ-IEC-021 item 4).
Dates	available by 2028-08-09; needed by 2028-08-16
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0008
Checks (engine)	OK completeness: electrical: complete OK short circuit: prospective 169.0 kA <= rating 200.0 kA INFO dates: available 2028-08-09, needed 2028-08-16: float 7 d (tight)

IF-37 MV supplies

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / electrical
Medium / location	electrical / SFC isolation transformer, IEC MV consumers
Size / connection	10.5 kV feeders / cable terminations at the IEC equipment (lugs by EPC)
Scope break	EPC 10.5 kV feeders and cables up to and including the terminations at the IEC equipment.
Istanbul EPC side	EPC: 10.5 kV switchgear feeders (EPC)
Imaginary Electric side	IEC: SFC isolation transformer, IEC MV consumers
Isolation	EPC feeder circuit breaker with earthing switch.
Protection	EPC feeder protection.
Flow	normal - / max 10.00 MVA
Electrical	10.50 kV (9.4-11.6 kV); Isc 35.0 kA, rating 40.0 kA
Conditions (text)	SFC 8.5 MW for 30 min at start-up.
Dates	available by 2028-08-09; needed by 2028-08-16
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0008
Checks (engine)	OK completeness: electrical: complete OK short circuit: prospective 35.0 kA <= rating 40.0 kA INFO dates: available 2028-08-09, needed 2028-08-16: float 7 d (tight)

IF-38 LV supplies

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / electrical
Medium / location	electrical / IEC MCCs incoming
Size / connection	0.4 kV feeders / IEC MCC incoming terminals
Scope break	EPC 0.4 kV feeders and cables up to the IEC MCC incoming terminals.
Istanbul EPC side	EPC: 0.4 kV switchgear feeders (EPC)
Imaginary Electric side	IEC: IEC MCCs incoming
Isolation	EPC feeder breaker; IEC MCC incomer.
Protection	EPC feeder protection selective with IEC MCC.
Electrical	0.40 kV (0.4-0.4 kV); Isc 50.0 kA, rating 50.0 kA
Conditions (text)	Loads per IEC-ALP-REQ-001 section 7.1.
Dates	available by 2028-08-09; needed by 2028-08-16
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0008
Checks (engine)	OK completeness: electrical: complete OK short circuit: prospective 50.0 kA <= rating 50.0 kA INFO dates: available 2028-08-09, needed 2028-08-16: float 7 d (tight)

IF-39 DC and UPS

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / electrical
Medium / location	electrical / TCS, protection, emergency oil pumps
Size / connection	220 V DC / 230 V AC UPS / IEC panel incoming terminals
Scope break	EPC DC and UPS distribution feeders up to the IEC panel terminals.
Istanbul EPC side	EPC: 220 V DC and 230 V AC UPS (EPC)
Imaginary Electric side	IEC: TCS, protection, emergency oil pumps
Isolation	EPC feeder MCB/fuse.
Protection	EPC selective fusing.
Electrical	0.22 kV (0.2-0.2 kV); Isc - kA, rating - kA

Conditions (text)	GT 37 kW / ST 45 kW DC emergency oil pumps, 3 h autonomy; 45 kVA UPS for the TCS.
Dates	available by 2028-08-09; needed by 2028-08-16
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0008
Checks (engine)	OK completeness: electrical: complete INFO dates: available 2028-08-09, needed 2028-08-16: float 7 d (tight)

IF-40 Control system interface

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / control
Medium / location	data / IEC turbine control system (TCS) gateway
Size / connection	OPC UA + Modbus TCP, redundant; hardwired trips / OPC UA + Modbus TCP, redundant fibre; hardwired trips
Scope break	IEC TCS gateway ports and fibre patch panel in the IEC electronic room; EPC fibre cables, DCS and the firewall between the zones (IEC 62443 zone/conduit).
Istanbul EPC side	EPC: plant DCS (EPC)
Imaginary Electric side	IEC: IEC turbine control system (TCS) gateway
Isolation	-
Protection	-
Conditions (text)	About 2,500 signals.
Dates	available by 2028-06-01; needed by 2028-06-15
Scope items	SC-013
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0008
Checks (engine)	OK completeness: control: complete OK dates: available 2028-06-01, needed 2028-06-15: float 14 d

IF-41 Protection interface

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / control
Medium / location	signals / generator/GSU protection panels
Size / connection	IEC 61850 + hardwired / IEC 61850 (GOOSE/MMS) + hardwired trips
Scope break	IEC generator/GSU protection panel terminals; EPC switchyard protection, substation automation and TSO SCADA.
Istanbul EPC side	EPC: switchyard protection and TSO SCADA (EPC)
Imaginary Electric side	IEC: generator/GSU protection panels
Isolation	-
Protection	-
Dates	available by 2028-07-29; needed by 2028-08-02
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0008
Checks (engine)	OK completeness: control: complete INFO dates: available 2028-07-29, needed 2028-08-02: float 4 d (tight)

IF-42 Earthing

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / civil
Medium / location	electrical / equipment earthing bosses
Size / connection	- / bolted earthing pads (IEC), EPC 240 mm ² Cu risers
Scope break	IEC equipment earthing pads and internal earthing; EPC earthing grid (IEEE 80) and risers.
Istanbul EPC side	EPC: earthing grid (EPC)
Imaginary Electric side	IEC: equipment earthing bosses
Isolation	-
Protection	-
Dates	available by 2027-10-15; needed by 2027-11-15
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0008
Checks (engine)	OK completeness: civil: complete OK dates: available 2027-10-15, needed 2027-11-15: float 31 d

IF-43 Foundations

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / civil
Medium / location	civil / anchor bolts, sole plates, load data (IEC)
Size / connection	- / anchor bolts / sole plates (IEC supply), grout by EPC to IEC specification
Scope break	IEC load data, anchor bolts, sole plates and grouting specification; EPC foundation design, construction, cast-in of anchor bolts and grouting.
Istanbul EPC side	EPC: foundation design and construction (EPC)
Imaginary Electric side	IEC: anchor bolts, sole plates, load data (IEC)
Isolation	-

Protection	-
Conditions (text)	Loads per IEC-ALP-REQ-001 section 9.
Dates	available by 2027-11-01; needed by 2027-11-15
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0008
Checks (engine)	OK completeness: civil: complete OK dates: available 2027-11-01, needed 2027-11-15: float 14 d

IF-44 Condenser tube cleaning balls

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / process
Medium / location	seawater + sponge balls / condenser inlet water boxes (injection) and outlet water box flanges (strainers)
Size / connection	2 x DN150 per inlet half / EN 1092-1 PN10 flanges
Scope break	IEC ball injection nozzles on the condenser inlet water boxes; EPC tube cleaning system (ball strainers in the outlet lines, recirculation pumps, collectors).
Istanbul EPC side	EPC: ball recirculation units, strainers, debris filters, controls
Imaginary Electric side	IEC: injection nozzles and outlet flanges on the water boxes
Isolation	EPC valves at the nozzles.
Protection	As IF-27.
Design	4.0 barg / 40 degC
Operating envelope	0.500 to 3.5 barg; 5 to 35 degC
Flow	normal - / max 200.00 m3/h (density 1025.0 kg/m3)
Quality / specification	source:SRC-OWNR-0002, requirement:ER-02.04
Dates	available by 2028-09-01; needed by 2028-09-15
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	decision:DEC-EPCE-0004, source:SRC-IECE-0008
Remarks	added by TQ-IEC-005
Checks (engine)	OK completeness: process: complete OK dates: available 2028-09-01, needed 2028-09-15: float 14 d

IF-45 HP attemperation spray water

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / process
Medium / location	feedwater / HRSG HP attemperator spray control station inlet
Size / connection	3 in (DN80) CL2500 / butt-weld end
Scope break	IEC HP attemperator spray control station inlet (station with control and block valves by IEC); EPC spray water line from the BFP discharge and the weld.
Istanbul EPC side	EPC: spray water line from the BFP discharge
Imaginary Electric side	IEC: HP attemperator spray control station
Isolation	EPC isolation valve at the BFP discharge header; IEC block valve.
Protection	As IF-18 (BFP shut-off <= 234 barg).
Design	234.0 barg / 200 degC
Operating envelope	95.0 to 215.0 barg; 130 to 170 degC
Flow	normal - / max 5.00 kg/s
HMB envelope (all cases)	stream 29 (full check): 1.39-2.06 kg/s, 143.744-194.605 barg, 147.6-158.7 degC
Quality / specification	requirement:ER-09.03, source:SRC-EPCE-0010
Dates	available by 2028-09-01; needed by 2028-09-15
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	source:SRC-IECE-0008, source:SRC-IECE-0005, decision:DEC-EPCE-0004
Checks (engine)	OK completeness: process: complete OK HMB flow: stream 29 max 2.06 kg/s (MAX-NG-100) <= capacity 5.00 kg/s (margin +58.8 %), 4 cases OK HMB pressure: stream 29: 143.744 - 194.605 barg within 95.0 - 215.0 barg (design 234.0) OK HMB temperature: stream 29: 147.6 - 158.7 degC within 130.0 - 170.0 degC (design 200.0) OK dates: available 2028-09-01, needed 2028-09-15: float 14 d

IF-46 Reheat attemperation spray water

Contract / category	ALP-CA-001 Consortium agreement IEPC - IEC: power island supply and division of responsibility / process
Medium / location	feedwater / HRSG reheat attemperator spray control station inlet
Size / connection	2 in (DN50) CL600 / butt-weld end
Scope break	IEC reheat attemperator spray control station inlet; EPC spray water line from the BFP interstage.
Istanbul EPC side	EPC: spray water line from the BFP interstage
Imaginary Electric side	IEC: reheat attemperator spray control station
Isolation	EPC isolation valve at the interstage header; IEC block valve.
Protection	As IF-19 (interstage <= 58 bar(a)).
Design	59.0 barg / 200 degC
Operating envelope	20.0 to 48.0 barg; 130 to 170 degC
Flow	normal - / max 3.00 kg/s
HMB envelope (all cases)	stream 30 (full check): 0.30-0.95 kg/s, 31.811-41.799 barg, 145.7-155.9 degC
Quality / specification	requirement:ER-09.03, source:SRC-EPCE-0010

Dates	available by 2028-09-01; needed by 2028-09-15
Agreement	agreed by Istanbul EPC, Imaginary Electric in TQ-IEC-021 (2026-09-28)
Basis	source:SRC-IECE-0008, source:SRC-IECE-0005, decision:DEC-EPCE-0004
Checks (engine)	OK completeness: process: complete OK HMB flow: stream 30 max 0.95 kg/s (MAX-NG-100) <= capacity 3.00 kg/s (margin +68.4 %), 4 cases OK HMB pressure: stream 30: 31.811 - 41.799 barg within 20.0 - 48.0 barg (design 59.0) OK HMB temperature: stream 30: 145.7 - 155.9 degC within 130.0 - 170.0 degC (design 200.0) OK dates: available 2028-09-01, needed 2028-09-15: float 14 d

4. Check results

WARN = to be resolved before the point is frozen; INFO = noted; OK rows are listed in the Excel register.

Point	Check	Result	Detail
IF-03	HMB scope	INFO	stream 28: flow + temperature checked only - see remarks
IF-28	dates	INFO	available 2028-08-09, needed 2028-08-16: float 7 d (tight)
IF-29	dates	INFO	available 2028-08-09, needed 2028-08-16: float 7 d (tight)
IF-31	dates	INFO	available 2028-08-25, needed 2028-09-01: float 7 d (tight)
IF-33	dates	INFO	available 2028-08-17, needed 2028-08-24: float 7 d (tight)
IF-34	dates	INFO	available 2028-07-29, needed 2028-08-02: float 4 d (tight)
IF-35	dates	INFO	available 2028-07-29, needed 2028-08-02: float 4 d (tight)
IF-36	dates	INFO	available 2028-08-09, needed 2028-08-16: float 7 d (tight)
IF-37	dates	INFO	available 2028-08-09, needed 2028-08-16: float 7 d (tight)
IF-38	dates	INFO	available 2028-08-09, needed 2028-08-16: float 7 d (tight)
IF-39	dates	INFO	available 2028-08-09, needed 2028-08-16: float 7 d (tight)
IF-41	dates	INFO	available 2028-07-29, needed 2028-08-02: float 4 d (tight)
TP-CW1	HMB scope	INFO	stream 25: flow + temperature checked only - see remarks
TP-CW2	HMB scope	INFO	stream 26: flow + temperature checked only - see remarks
TP-E1	HMB power	INFO	max net 665.7 MW (WIN-NG-100) > 640 MW (margin -4.0 %); limited to 640 MW at this point: Export limiter in the plant control system: 640 MW net at the 380 kV revenue meters (TQ-027); line differential 87L main 1 / main 2 with the remote end (plant end by the Contractor, remote end by the Owner), settings coordinated with the TSO (ER-18.02).
TP-E1	dates	INFO	available 2028-07-15, needed 2028-07-22: float 7 d (tight)
-	HMB coverage	INFO	streams not mapped to a tie-in (internal to one scope, else an interface is missing): 1 Ambient air to GT compressor inlet; 3 Heated fuel gas to GT combustion system; 4 GT exhaust gas to HRSG inlet; 10 IP superheated steam (joins cold reheat inside the HRSG); 15 IP turbine exhaust (crossover, internal); 16 LP turbine exhaust to condenser (UEEP); 18 Condensate at CEP discharge; 24 IP economiser water to performance fuel gas heater
-	coverage	INFO	SC-006 terminal point 'GT exhaust flange to HRSG inlet duct' not referenced by a tie-in
-	coverage	INFO	SC-007 terminal point 'Generator terminals (IPB connection)' not referenced by a tie-in
-	coverage	INFO	SC-009 terminal point 'Generator terminals' not referenced by a tie-in
-	coverage	INFO	SC-051 terminal point 'Owner's warehouse' not referenced by a tie-in